Chapter 4

Phase Shift Keying

Phase shift keying (PSK) is a large class of digital modulation schemes. PSK is widely used in the communication industry. In this chapter we study each PSK modulation scheme in a single section where signal description, power spectral density, modulator/demodulator block diagrams, and receiver error performance are all included. First we present coherent binary PSK(BPSK) and its noncoherent counterpart, differential BPSK (DBPSK), in Sections 4.1 and 4.2. Then we discuss in Section 4.3 M-ary PSK (MPSK) and its PSD in Section 4.4. The noncoherent version, differential MPSK (DMPSK) is treated in Section 4.5. We discuss in great detail quadrature PSK (QPSK) and differential QPSK (DQPSK) in Sections 4.6 and 4.7, respectively. Section 4.8 is a brief discussion of offset QPSK (OQPSK). An important variation of QPSK, the $\pi/4$–DQPSK which has been designated as the American standard of the second-generation cellular mobile communications, is given in Section 4.9. Section 4.10 is devoted to carrier and clock recovery. Finally, we summarize the chapter with Section 4.11.

4.1 BINARY PSK

Binary data are represented by two signals with different phases in BPSK. Typically these two phases are 0 and π, the signals are

$$\begin{aligned} s_1(t) &= A\cos 2\pi f_c t, \quad 0 \le t \le T, \quad \text{for } 1 \\ s_2(t) &= -A\cos 2\pi f_c t, \quad 0 \le t \le T, \quad \text{for } 0 \end{aligned} \tag{4.1}$$

These signals are called *antipodal*. The reason that they are chosen is that they have a correlation coefficient of -1, which leads to the minimum error probability for the same E_b/N_o, as we will see shortly. These two signals have the same frequency and energy.

As we will see in later sections, all PSK signals can be graphically represented

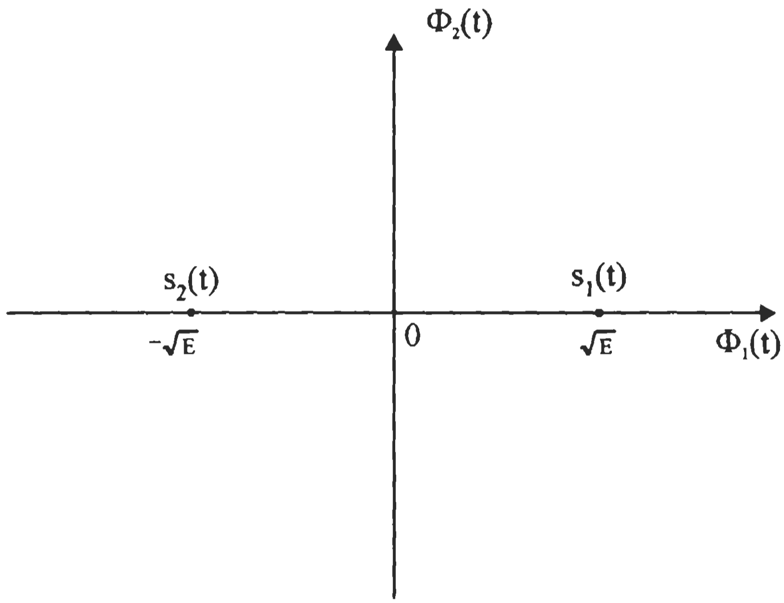


Figure 4.1 BPSK signal constellation.

by a *signal constellation* in a two-dimensional coordinate system with

$$\phi_1(t) = \sqrt{\frac{2}{T}} \cos 2\pi f_c t, \qquad 0 \le t \le T \tag{4.2}$$

and

$$\phi_2(t) = -\sqrt{\frac{2}{T}} \sin 2\pi f_c t, \qquad 0 \le t \le T \tag{4.3}$$

as its horizontal and vertical axis, respectively. Note that we deliberately add a minus sign in $\phi_2(t)$ so that PSK signal expressions will be a sum instead of a difference (see (4.14)). Many other signals, especially QAM signals, can also be represented in the same way. Therefore we introduce the signal constellation of BPSK here as shown in Figure 4.1 where $s_1(t)$ and $s_2(t)$ are represented by two points on the horizontal axis, respectively, where

$$E = \frac{A^2 T}{2}$$

The waveform of a BPSK signal generated by the modulator in Figure 4.3 for a data stream $\{10110\}$ is shown in Figure 4.2. The waveform has a constant envelope like FSK. Its frequency is constant too. In general the phase is not continuous at

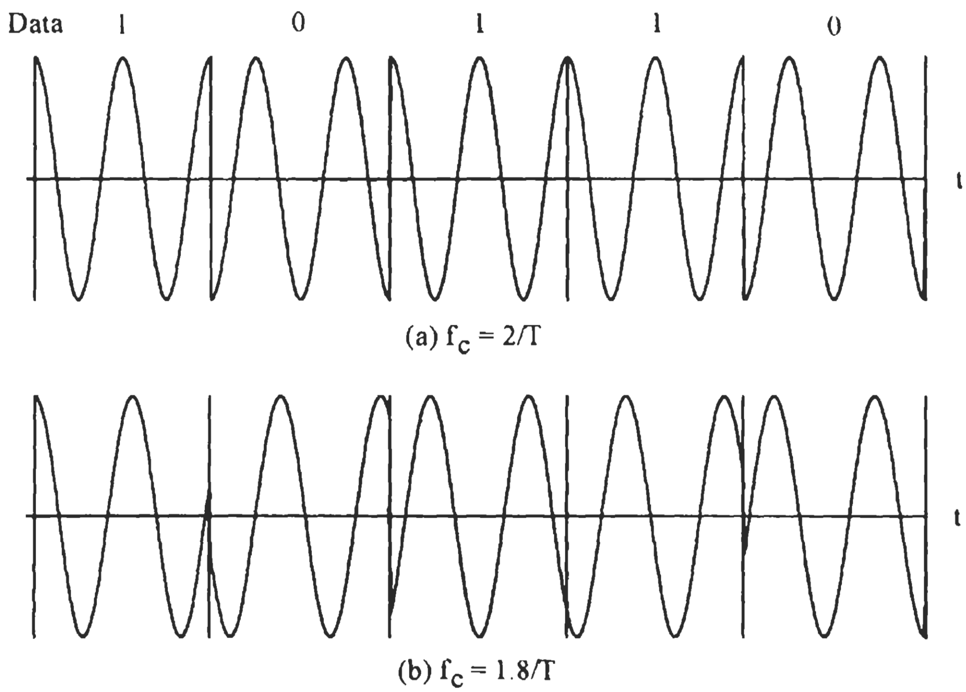


Figure 4.2 BPSK waveforms.

bit boundaries. If the $f_c = m\ R_b = m/T$, where m is an integer and R_b is the data bit rate, and the bit timing is synchronous with the carrier, then the initial phase at a bit boundary is either 0 or π (Figure 4.2(a)), corresponding to data bit 1 or 0. However, if the f_c is not an integer multiple of R_b , the initial phase at a bit boundary is neither 0 nor π (Figure 4.2(b)). In other words, the modulated signals are not the ones given in (4.1). We will show next in discussion of demodulation that condition $f_c = m\ R_b$ is necessary to ensure minimum bit error probability. However, if $f_c >> R_b$, this condition can be relaxed and the resultant BER performance degradation is negligible.[1]

The modulator which generates the BPSK signal is quite simple (Figure 4.3 (a)). First a bipolar data stream $a(t)$ is formed from the binary data stream

$$a(t) = \sum_{k=-\infty}^{\infty} a_k p(t - kT) \tag{4.4}$$

[1] This is true for all PSK schemes and PSK-derived schemes, including QPSK, MSK, and MPSK. We will not mention this again when we discuss other PSK schemes.

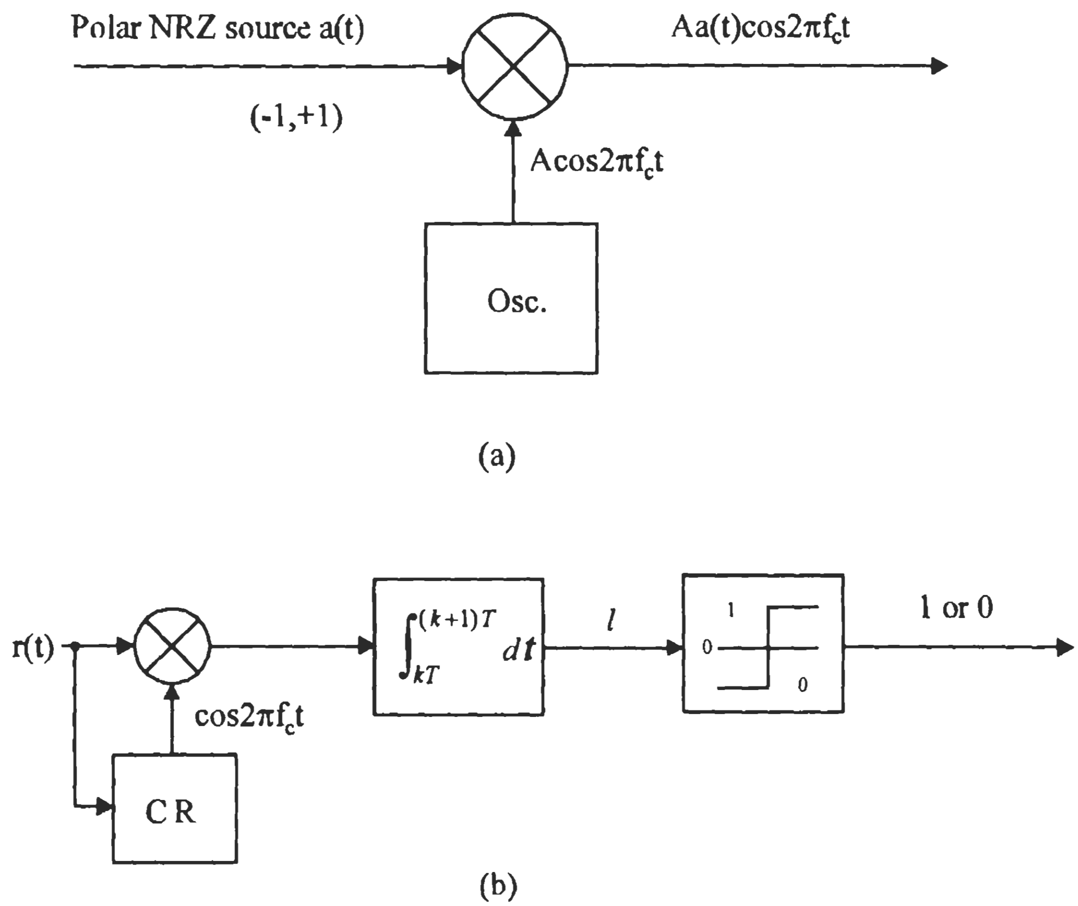


Figure 4.3 BPSK modulator (a), and coherent BPSK demodulator (b).

where $a_k \in \{+1, -1\}$, $p(t)$ is the rectangular pulse with unit amplitude defined on $[0, T]$. Then $a(t)$ is multiplied with a sinusoidal carrier $A \cos 2\pi f_c t$. The result is the BPSK signal

$$s(t) = Aa(t) \cos 2\pi f_c t, \quad -\infty < t < \infty \tag{4.5}$$

Note that the bit timing is not necessarily synchronous with the carrier.

The coherent demodulator of BPSK falls in the class of coherent detectors for binary signals as described in Appendix B. The coherent detector could be in the form of a correlator or matched filter. The correlator's reference signal is the difference signal ($s_d(t) = 2A \cos 2\pi f_c t$). Figure 4.3(b) is the coherent receiver using a correlator where the reference signal is the scaled-down version of the difference signal. The reference signal must be synchronous to the received signal in frequency and phase.

It is generated by the carrier recovery (CR) circuit. Using a matched filter instead of a correlator is not recommended at passband since a filter with $h(t) = \cos 2\pi f_c(T-t)$ is difficult to implement.

In the absence of noise, setting $A = 1$, the output of the correlator at $t = (k+1)T$ is

$$\begin{aligned}
&\int_{kT}^{(k+1)T} r(t)\cos 2\pi f_c t dt \\
=&\int_{kT}^{(k+1)T} a_k \cos^2 2\pi f_c t dt \\
=&\frac{1}{2}\int_{kT}^{(k+1)T} a_k(1+\cos 4\pi f_c t)dt \\
=&\frac{T}{2}a_k + \frac{a_k}{8\pi f_c}\left[\sin 4\pi f_c(k+1)T - \sin 4\pi f_c kT\right]
\end{aligned}$$

If $f_c = m\, R_b$, the second term is zero, thus the original signal $a(t)$ is perfectly recovered (in the absence of noise). If $f_c \neq m\, R_b$, the second term will not be zero. However, as long as $f_c >> R_b$, the second term is much smaller than the first term so that its effect is negligible.

The bit error probability can be derived from the formula for general binary signals (Appendix B):

$$P_b = Q\left(\sqrt{\frac{E_1 + E_2 - 2\rho_{12}\sqrt{E_2E_1}}{2N_o}}\right)$$

For BPSK $\rho_{12} = -1$ and $E_1 = E_2 = E_b$, thus

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_o}}\right), \text{ (coherent BPSK)} \tag{4.6}$$

A typical example is that, at $E_b/N_o = 9.6$ dB, $P_b = 10^{-5}$. Figure 4.4 shows the P_b curve of BPSK. The curves of coherent and noncoherent BFSK are also shown in the figure. Recall the P_b expression for coherent BFSK is $P_b = Q\left(\sqrt{\frac{E_b}{N_o}}\right)$ which is 3 dB inferior to coherent BPSK. However, coherent BPSK requires that the reference signal at the receiver to be synchronized in phase and frequency with the received signal. This will be discussed in Section 4.10. Noncoherent detection of BPSK is also possible. It is realized in the form of differential BPSK which will be studied in the next section.

Next we proceed to find the power spectral density of the BPSK signal. It suf-

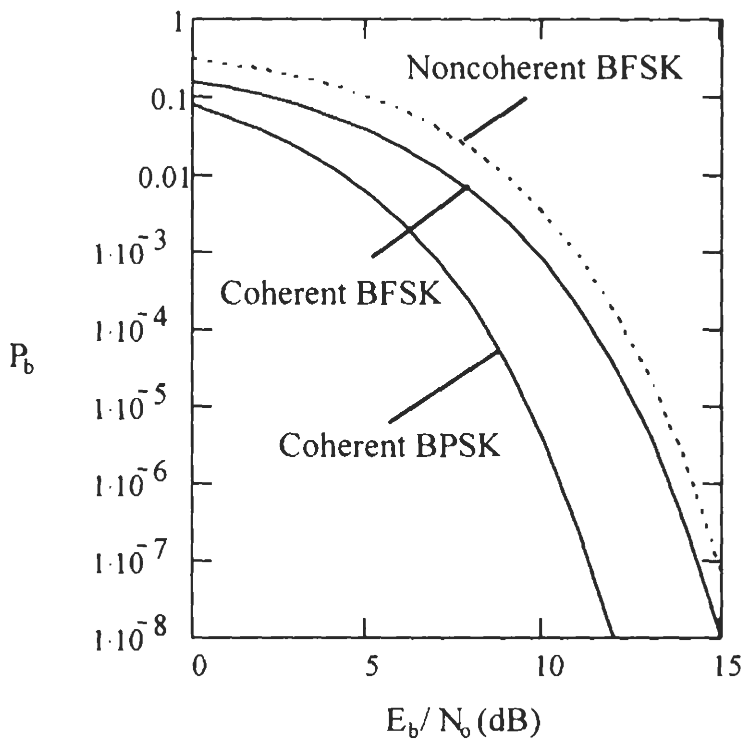


Figure 4.4 P_b of BPSK in comparison with BFSK.

fices to find the PSD of the baseband shaping pulse. As shown in Appendix A, the PSD of a binary, bipolar, equiprobable, stationary, and uncorrelated digital waveform is just equal to the energy spectral density of the pulse divided by the symbol duration (see (A.19)). The basic pulse of BPSK is just a rectangular pulse[2]

$$p(t) = \begin{cases} A, & 0 < t < T \\ 0, & \text{otherwise} \end{cases} \tag{4.7}$$

Its Fourier transform is

$$G(f) = AT\frac{\sin \pi fT}{\pi fT}e^{-2\pi fT/2}$$

Thus the PSD of the baseband BPSK signal is

$$\Psi_{\tilde{s}}(f) = \frac{|G(f)|^2}{T} = A^2T\left(\frac{\sin \pi fT}{\pi fT}\right)^2, \text{ (BPSK)} \tag{4.8}$$

[2] The bipolarity of the baseband waveform of BPSK is controlled by the bipolar data $a_k = \pm 1$.

which is plotted in Figure 4.5. From the figure we can see that the null-to-null bandwidth $B_{null} = 2/T = 2R_b$. (Keep in mind that the PSD at the carrier frequency is two-sided about f_c.) Figure 4.5(c) is the out-of-band power curve which is defined by (2.21). From this curve we can estimate that $B_{90\%} \approx 1.7R_b$ (corresponding to -10 dB point on the curve). We also calculated that $B_{99\%} \approx 20R_b$.

4.2 DIFFERENTIAL BPSK

In Chapter 2 we first introduced differential encoding and decoding of binary data. This technique can be used in PSK modulation. We denote differentially encoded BPSK as DEBPSK. Figure 4.6 (a) is the DEBPSK modulator. DEBPSK signal can be coherently demodulated or differentially demodulated. We denote the modulation scheme that uses differential encoding and differential demodulation as DBPSK, which is sometimes simply called DPSK.

DBPSK does not require a coherent reference signal. Figure 4.6(b) is a simple, but suboptimum, differential demodulator which uses the previous symbol as the reference for demodulating the current symbol.[3] The front-end bandpass filter reduces noise power but preserves the phase of the signal. The integrator can be replaced by an LPF. The output of the integrator is

$$l = \int_{kT}^{(k+1)T} r(t)r(t-T)dt$$

In the absence of noise and other channel impairment,

$$l = \int_{kT}^{(k+1)T} s_k(t)s_{k-1}(t)dt = \begin{cases} E_b, & \text{if } s_k(t) = s_{k-1}(t) \\ -E_b, & \text{if } s_k(t) = -s_{k-1}(t) \end{cases}$$

where $s_k(t)$ and $s_{k-1}(t)$ are the current and the previous symbols. The integrator output is positive if the current signal is the same as the previous one, otherwise the output is negative. This is to say that the demodulator makes decisions based on the difference between the two signals. Thus information data must be encoded as the difference between adjacent signals, which is exactly what the differential encoding can accomplish. Table 4.1 shows an example of differential encoding, where an arbitrary reference bit 1 is chosen. The encoding rule is

$$d_k = \overline{a_k \oplus d_{k-1}}$$

[3] This is the commonly referred DPSK demodulator. Another DPSK demodulator is the optimum differentially coherent demodulator. Differentially encoded PSK can also be coherently detected. These will be discussed shortly.

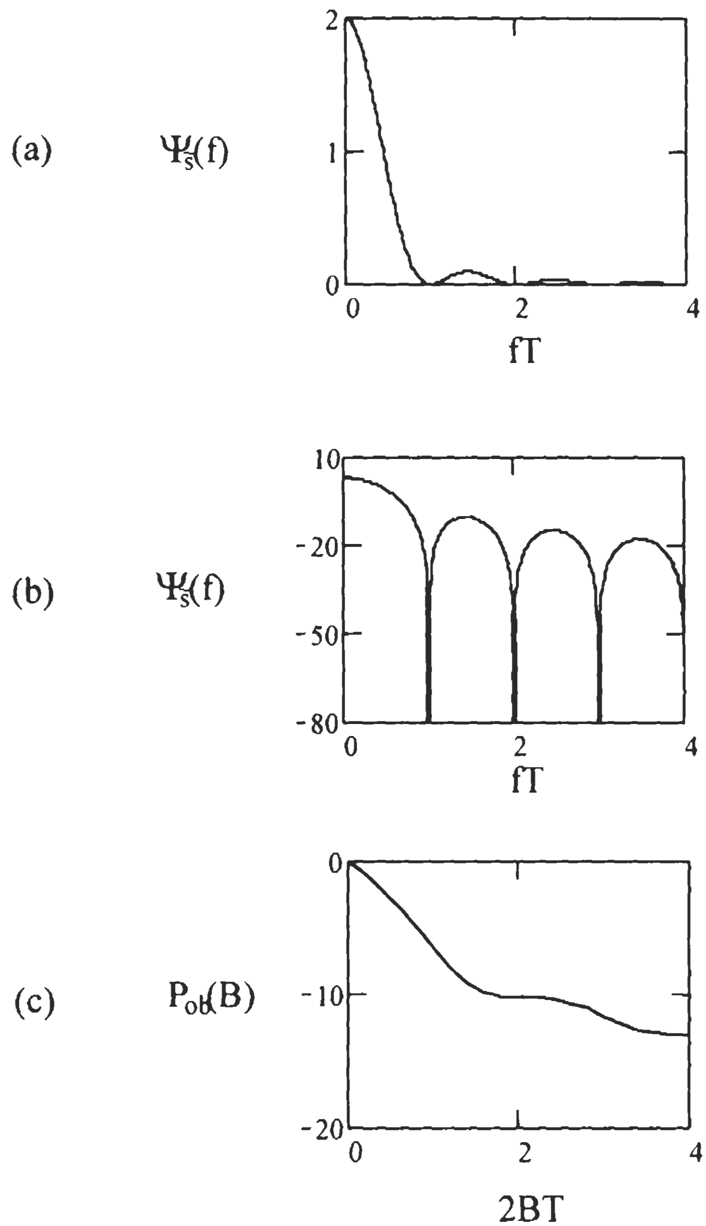


Figure 4.5 Power spectral density of BPSK: (a) linear, (b) logarithmic, (c) out-of-band power.

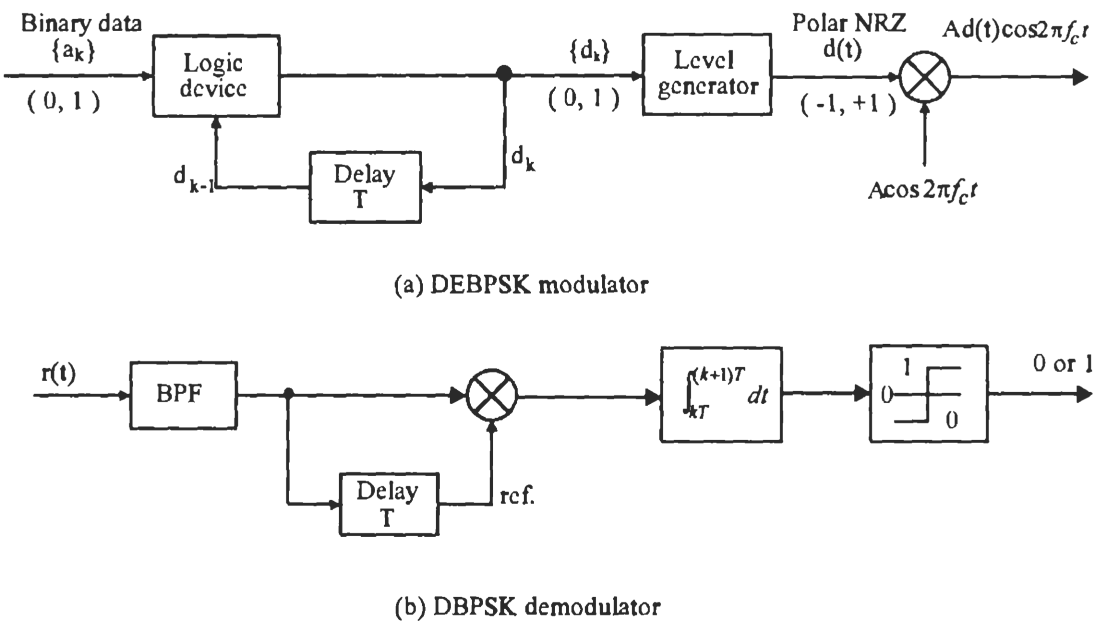


Figure 4.6 DBPSK modulator (a), and demodulator (b).

Inversely we can recover a_k from d_k using

$$a_k = \overline{d_k \oplus d_{k-1}}$$

If d_k and d_{k-1} are the same, then they represent a 1 of a_k. If d_k and d_{k-1} are different, they represent a 0 of a_k. This can be verified by comparing sequences $\{d_k\}$ and $\{a_k\}$ in the table. The sequence $\{d_k\}$ is modulated onto a carrier with phase 0 or π. In the absence of noise and other channel impairment, the demodulator output $\widehat{a}_k$ is identical to the message sequence.

The preceding receiver is suboptimum, since the reference signal is the previous symbol which is noisy. The optimum noncoherent, or differentially coherent, demodulation of DEBPSK is presented now. As discussed above, a message bit is represented by two modulated symbols. If the transmitted bit is 1, the two symbols are the same. Thus we can define a signal with a duration of $2T$ as follows to represent binary 1

$$\xi_1(t) = \begin{cases} A\cos 2\pi f_c t, & 0 \le t \le T \\ A\cos 2\pi f_c t, & T \le t \le 2T \end{cases}, \quad \text{for binary 1}$$

Modulation	ref.									
Message a_k		1	0	1	1	0	0	0	1	1
Encoding $d_k = \overline{a_k \oplus d_{k-1}}$	1	1	0	0	0	1	0	1	1	1
Signal phase θ	0	0	π	π	π	0	π	0	0	0
Demodulation										
$\frac{l}{E_b} = \frac{1}{E_b}\int_{kT}^{(k+1)T} s_k(t)s_{k-1}(t)dt$		1	-1	1	1	-1	-1	-1	1	1
Demodulator output $\widehat{a}_k$		1	0	1	1	0	0	0	1	1

Table 4.1 Examples of differential coding.

If the transmitted bit is 0, the two symbols are different. Thus we can define

$$\xi_2(t) = \begin{cases} A\cos 2\pi f_c t, & 0 \le t \le T \\ -A\cos 2\pi f_c t, & T \le t \le 2T \end{cases}, \quad \text{for binary 0}$$

Note that in the modulated signal stream, the $2T$-symbols are overlapped by T seconds.

Since we desire an optimum noncoherent demodulation, the DBPSK receiver may be implemented in the general forms for signals with unknown phases as depicted in Appendix B. However, a simpler form is possible due to the special property of the signals. The simpler form avoids the squarers or matched filters. We derive this receiver starting from (B.55). Assuming the received signal is $r(t)$, the sufficient statistic for $\xi_1(t)$ is

$$\begin{aligned} l_1^2 &= \left(\int_0^{2T} r(t)\xi_1(t)dt\right)^2 + \left(\int_0^{2T} r(t)\xi_1(t, \frac{\pi}{2})dt\right)^2 \\ &= \left(\int_0^{2T} r(t)A\cos 2\pi f_c tdt\right)^2 + \left(\int_0^{2T} r(t)A\sin 2\pi f_c tdt\right)^2 \\ &= (w_0 + w_1)^2 + (z_0 + z_1)^2 \end{aligned}$$

where

$$w_0 \triangleq \int_0^T r(t)A\cos 2\pi f_c tdt$$

$$w_1 \triangleq \int_T^{2T} r(t)A\cos 2\pi f_c tdt$$

$$z_0 \triangleq \int_0^T r(t)A\sin 2\pi f_c tdt$$

$$z_1 \triangleq \int_T^{2T} r(t) A \sin 2\pi f_c t dt$$

Similarly, the sufficient statistic for $\xi_2(t)$ is

$$l_2^2 = (w_0 - w_1)^2 + (z_0 - z_1)^2$$

The decision rule is

$$l_1^2 \underset{0}{\overset{1}{\gtrless}} l_2^2$$

Substituting expressions for l_1^2 and l_2^2 into the above expression and cancelling like terms, we obtain

$$x \triangleq w_1 w_0 + z_1 z_0 \underset{0}{\overset{1}{\gtrless}} 0$$

For the kth symbol period, this rule is

$$x_k \triangleq w_k w_{k-1} + z_k z_{k-1} \underset{0}{\overset{1}{\gtrless}} 0 \tag{4.9}$$

This rule can be implemented by the receiver shown in Figure 4.7. The reference signals are locally generated since phase synchronization between $r(t)$ and the reference signals is not required. However, the frequency of the reference signals must be the same as the received signal's. This can be maintained by using stable oscillators, such as crystal oscillators, in both transmitter and receiver. However, in the case where Doppler shift exists in the carrier frequency, such as in mobile communications, frequency tracking is needed to maintain the same frequency. In this case the local oscillator must be synchronized in frequency to the received signal. The reference signals' amplitude A is set as 1 in Figure 4.7. In fact A could be any value since its value will not affect the decision rule in (4.9). The correlators produce w_k and z_k. The x_k is calculated by the delay-and-multiply circuits or differential decoders.

To derive the error probability of the optimum demodulator, we observe that two DBPSK symbols are orthogonal over $[0, 2T]$ since

$$\int_0^{2T} \xi_1(t)\xi_2(t)dt = \int_0^T (A\cos 2\pi f_c t)^2 dt - \int_T^{2T} (A\cos 2\pi f_c t)^2 = 0$$

In other words DBPSK is a special case of noncoherent orthogonal modulation with $T_s = 2T$ and $E_s = 2E_b$. Hence using the result of Appendix B (B.56) we have the

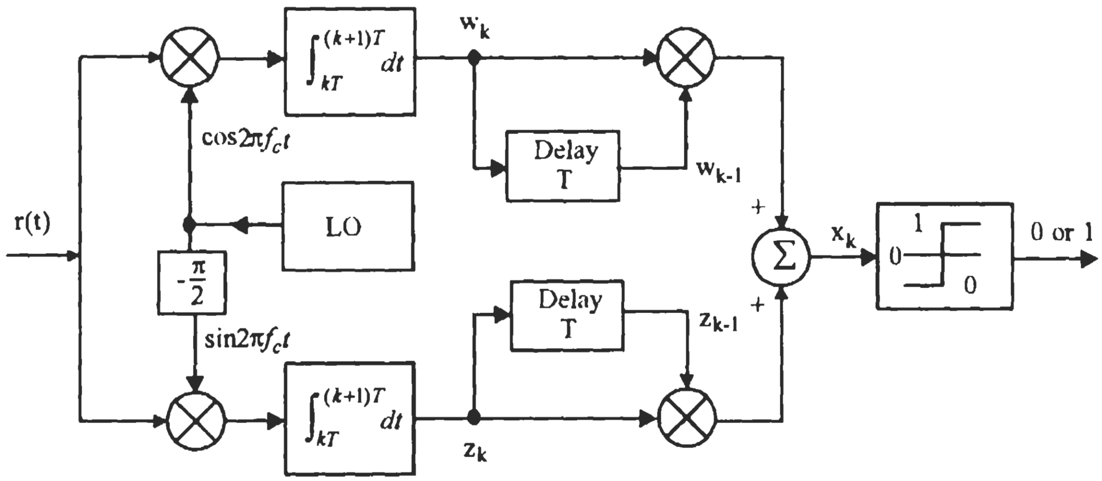


Figure 4.7 Optimum demodulator for DBPSK.

bit error probability

$$P_b = \frac{1}{2} e^{-E_b/N_o}, \text{ (optimum DBPSK)} \tag{4.10}$$

Note that the demodulator of Figure 4.7 does not require phase synchronization between the reference signals and the received signal. But it does require the reference frequency be the same as the received signal. Therefore the suboptimum receiver in Figure 4.6(b) is more practical, and indeed it is the usual-sense DBPSK receiver. Its error performance is slightly inferior to that of the optimum given in (4.10).

The performance of the suboptimum receiver is given by Park in [1]. It is shown that if an ideal narrow-band IF filter with bandwidth W is placed before the correlator in Figure 4.6(b), the bit error probability is

$$P_b = \frac{1}{2} e^{-0.76 E_b/N_o}, \quad \text{for } W = 0.5/T$$

or

$$P_b = \frac{1}{2} e^{-0.8 E_b/N_o}, \quad \text{for } W = 0.57/T$$

which amounts to a loss of 1.2 dB and 1 dB, respectively, with respect to the optimum.

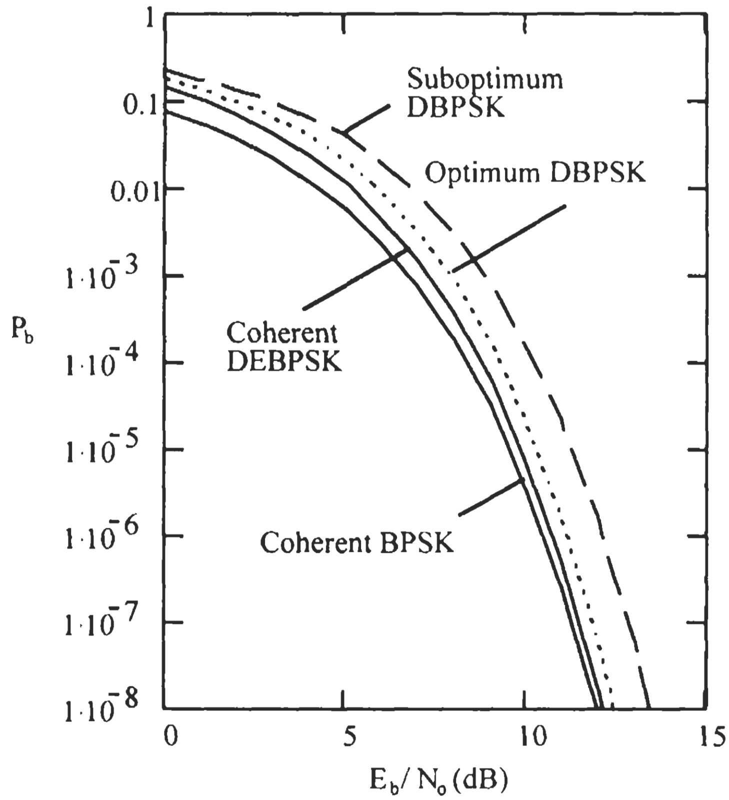


Figure 4.8 P_b of differential BPSK in comparison with coherent BPSK scheme.

If an ideal wide-band IF filter is used, then

$$
\begin{aligned}
P_b &\approx Q\left(\sqrt{\frac{E_b}{N_o}}\right), \quad \text{for } W > 1/T \\
&\approx \frac{1}{2\sqrt{\pi}\sqrt{E_b/2N_o}} e^{-E_b/2N_o}, \quad \text{for } W > 1/T \text{ , (Suboptimum DBPSK)}
\end{aligned}
$$

Typical value of W is 1.5/T. If W is too large or too small the above expression does not hold [1]. The P_b for the wide-band suboptimum receiver is about 2 dB worse than the optimum at high SNR. The bandwidth should be chosen as 0.57/T for the best performance. P_b curves of DBPSK are shown in Figure 4.8.

A differentially encoded BPSK signal can also be demodulated coherently (denoted as DEBPSK). It is used when the purpose of differential encoding is to eliminate phase ambiguity in the carrier recovery circuit for coherent PSK (see Section

4.10). This is not usually meant by the name DBPSK. DBPSK refers to the scheme of differential encoding and differentially coherent demodulation as we have discussed above.

In the case of DEBPSK, the bit error rate of the final decoded sequence $\{\widehat{a}_k\}$, P_b is related to the bit error rate of the demodulated encoded sequence $\{\widehat{d}_k\}$, $P_{b,d}$, by

$$P_b = 2P_{b,d}(1 - P_{b,d}) \tag{4.11}$$

as we have shown in Section 2.4.1 of Chapter 2. Substituting $P_{b,d}$ as in (4.6) into the above expression we have

$$P_b = 2Q\left(\sqrt{\frac{2E_b}{N_o}}\right)\left[1 - Q\left(\sqrt{\frac{2E_b}{N_o}}\right)\right], \text{ (DEBPSK)} \tag{4.12}$$

for coherently detected differentially encoded PSK. For large SNR, this is just about two times that of coherent BPSK without differential encoding.

Finally we need to say a few words of power spectral density of differentially encoded BPSK. Since the difference of differentially encoded BPSK from BPSK is differential encoding, which always produces an asymptotically equally likely data sequence (see Section 2.1), the PSD of the differentially encoded BPSK is the same as BPSK which we assume is equally likely. The PSD is shown in Figure 4.5. However, it is worthwhile to point out that if the data sequence is not equally likely the PSD of the BPSK is not the one in Figure 4.5, but the PSD of the differentially encoded PSK is still the one in Figure 4.5.

4.3 M-ARY PSK

The motivation behind MPSK is to increase the bandwidth efficiency of the PSK modulation schemes. In BPSK, a data bit is represented by a symbol. In MPSK, $n = \log_2 M$ data bits are represented by a symbol, thus the bandwidth efficiency is increased to n times. Among all MPSK schemes, QPSK is the most-often-used scheme since it does not suffer from BER degradation while the bandwidth efficiency is increased. We will see this in Section 4.6. Other MPSK schemes increase bandwidth efficiency at the expenses of BER performance.

M-ary PSK signal set is defined as

$$s_i(t) = A\cos(2\pi f_c t + \theta_i), \quad 0 \le t \le T, \quad i = 1, 2, \ldots, M \tag{4.13}$$

where

$$\theta_i = \frac{(2i-1)\pi}{M}$$

The carrier frequency is chosen as integer multiple of the symbol rate, therefore in any symbol interval, the signal initial phase is also one of the M phases. Usually M is chosen as a power of 2 (i.e., $M = 2^n$, $n = \log_2 M$). Therefore binary data stream is divided into n-tuples. Each of them is represented by a symbol with a particular initial phase.

The above expression can be written as

$$\begin{aligned} s_i(t) &= A\cos\theta_i \cos 2\pi f_c t - A\sin\theta_i \sin 2\pi f_c t \\ &= s_{i1}\phi_1(t) + s_{i2}\phi_2(t) \end{aligned} \tag{4.14}$$

where $\phi_1(t)$ and $\phi_2(t)$ are orthonormal basis functions (see (4.2) and (4.3)), and

$$s_{i1} = \int_0^T s_i(t)\phi_1(t)dt = \sqrt{E}\cos\theta_i$$

$$s_{i2} = \int_0^T s_i(t)\phi_2(t)dt = \sqrt{E}\sin\theta_i$$

where

$$E = \frac{1}{2}A^2T$$

is the symbol energy of the signal. The phase is related with s_{i1} and s_{i2} as

$$\theta_i = \tan\frac{s_{i2}}{s_{i1}}$$

The MPSK signal constellation is therefore two-dimensional. Each signal $s_i(t)$ is represented by a point (s_{i1}, s_{i2}) in the coordinates spanned by $\phi_1(t)$ and $\phi_2(t)$. The polar coordinates of the signal are $(\sqrt{E}, \theta_i)$. That is, its magnitude is $\sqrt{E}$ and its angle with respect to the horizontal axis is θ_i. The signal points are equally spaced on a circle of radius $\sqrt{E}$ and centered at the origin. The bits-signal mapping could be arbitrary provided that the mapping is one-to-one. However, a method called Gray coding is usually used in signal assignment in MPSK. Gray coding assigns n-tuples with only one-bit difference to two adjacent signals in the constellation. When an M-ary symbol error occurs, it is more likely that the signal is detected as the adjacent signal on the constellation, thus only one of the n input bits is in error. Figure 4.9 is the constellation of 8-PSK, where Gray coding is used for bit assignment. Note that

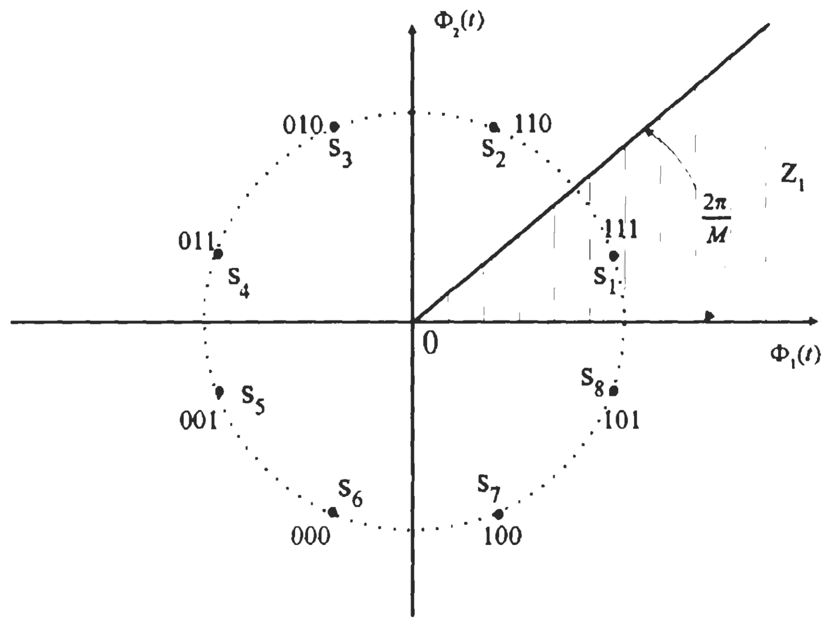


Figure 4.9 8-PSK constellation with Gray coded bit assignment.

BPSK and QPSK are special cases of MPSK with $M = 2$ and 4, respectively. On the entire time axis, we can write MPSK signal as

$$s(t) = s_1(t)\cos 2\pi f_c t - s_2(t)\sin 2\pi f_c t, \quad -\infty < t < \infty \tag{4.15}$$

where

$$s_1(t) = A \sum_{k=-\infty}^{\infty} \cos(\theta_k) p(t - kT) \tag{4.16}$$

$$s_2(t) = A \sum_{k=-\infty}^{\infty} \sin(\theta_k) p(t - kT) \tag{4.17}$$

where θ_k is one of the M phases determined by the input binary n-tuple, $p(t)$ is the rectangular pulse with unit amplitude defined on $[0, T]$. Expression (4.15) implies that the carrier frequency is an integer multiple of the symbol timing so that the initial phase of the signal in any symbol period is θ_k.

Since MPSK signals are two-dimensional, for $M \geq 4$, the modulator can be implemented by a quadrature modulator. The MPSK modulator is shown in Figure 4.10. The only difference for different values of M is the level generator. Each n-tuple of the input bits is used to control the level generator. It provides the I-

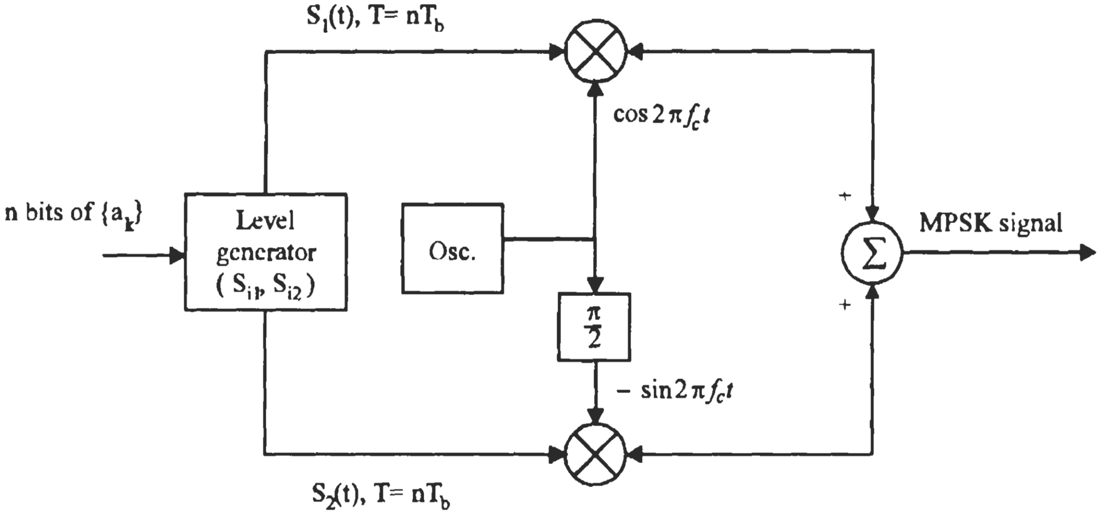


Figure 4.10 MPSK modulator.

and Q-channels the particular sign and level for a signal's horizontal and vertical coordinates, respectively. For QPSK, the level generator is particularly simple, it is simply a serial-to-parallel converter (see Section 4.6).

Modern technology intends to use completely digital devices. In such an environment, MPSK signals are digitally synthesized and fed to a D/A converter whose output is the desired phase modulated signal.

The coherent demodulation of MPSK could be implemented by one of the coherent detectors for M-ary signals as described in Appendix B. Since the MPSK signal set has only two basis functions, the simplest receiver is the one that uses two correlators (Figure B.8 with $N = 2$). Due to the special characteristic of the MPSK signal, the general demodulator of Figure B.8 can be further simplified. For MPSK the sufficient statistic is

$$\begin{aligned} l_i &= \int_0^T r(t)s_i(t)dt = \int_0^T r(t)[s_{i1}\phi_1(t) + s_{i2}\phi_2(t)]dt \\ &= \int_0^T r(t)[\sqrt{E}\cos\theta_i\phi_1(t) + \sqrt{E}\sin\theta_i\phi_2(t)]dt \\ &= \sqrt{E}\,[r_1\cos\theta_i + r_2\sin\theta_i] \end{aligned} \tag{4.18}$$

where

$$r_1 \triangleq \int_0^T r(t)\phi_1(t)dt = \int_0^T [s(t)+n(t)]\phi_1(t)dt = s_{i1} + n_1$$

$$r_2 \triangleq \int_0^T r(t)\phi_2(t)dt = \int_0^T [s(t)+n(t)]\phi_2(t)dt = s_{i2}+n_2$$

are independent Gaussian random variables with mean values s_{i1} and s_{i2}, respectively. Their variance is $N_o/2$.

Let

$$r_1 = \rho\cos\widehat{\theta}$$

$$r_2 = \rho\sin\widehat{\theta}$$

then

$$\rho = \sqrt{r_1^2 + r_2^2} \tag{4.19}$$

$$\widehat{\theta} \triangleq \tan^{-1}\frac{r_2}{r_1} \tag{4.20}$$

$$\begin{aligned} l_i &= \sqrt{E}[\rho\cos\widehat{\theta}\cos\theta_i + \rho\sin\widehat{\theta}\sin\theta_i] \\ &= \sqrt{E}\rho\cos(\theta_i - \widehat{\theta}) \end{aligned}$$

In the absence of noise, $\widehat{\theta} = \tan^{-1} r_2/r_1 = \tan^{-1} s_{i2}/s_{i1} = \theta_i$. With noise, $\widehat{\theta}$ will deviate from θ_i. Since ρ is independent of any signal, then choosing the largest l_i is equivalent to choosing the smallest $|\theta_i - \widehat{\theta}|$. This rule is in fact to choose signal $s_i(t)$ when $\mathbf{r} = \left[\begin{smallmatrix} r_1 \\ r_2 \end{smallmatrix}\right]$ falls inside the pie-shape decision region of the signal (see Figure 4.9). Figure 4.11 is the demodulator based on the above decision rule where subscript k indicates the kth symbol period and CR stands for carrier recovery. Note that the amplitude of the reference signals can be any value, which is $\sqrt{2/T}$ in the figure, since the effect of the amplitude is cancelled when computing $\widehat{\theta}_k$.

The symbol error probability can be derived as follows. Given $s_i(t)$ is transmitted (or hypothesis H_i is true), the received vector $\mathbf{r} = \left[\begin{smallmatrix} r_1 \\ r_2 \end{smallmatrix}\right]$ is a point in the $\phi_1(t) - \phi_2(t)$ plane. Its joint probability density function is two-dimensional.

$$p(\mathbf{r}/H_i) = \frac{1}{\pi N_o}\exp\{-\frac{1}{N_o}[(r_1 - \sqrt{E}\cos\theta_i)^2 + (r_2 - \sqrt{E}\sin\theta_i)^2]\}$$

Geometrically, the PDF is a bell-shape surface centered at $\mathbf{s}_i = \left[\begin{smallmatrix} s_{i1} \\ s_{i2} \end{smallmatrix}\right]$ (Figure 4.12).

An error occurs when $\mathbf{r}$ falls outside the decision region Z_i (see Figure 4.9). Thus

$$P_s = 1 - \int_{Z_i} p(\mathbf{r}/H_i)d\mathbf{r}$$

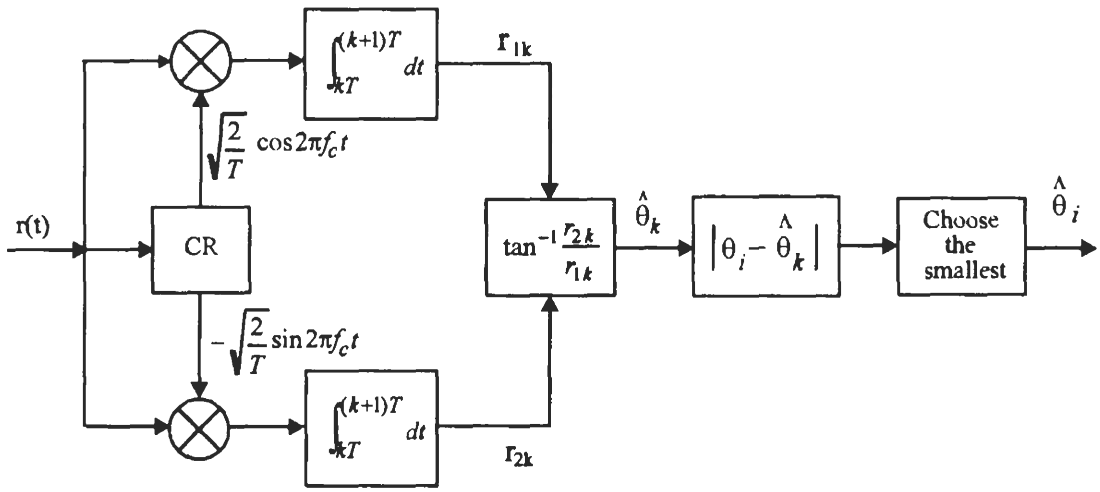


Figure 4.11 Coherent MPSK demodulator using two correlators.

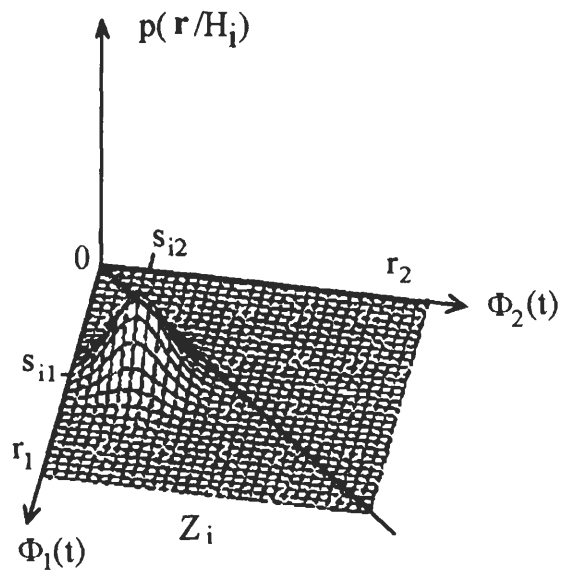


Figure 4.12 Joint PDF of **r** given $s_i(t)$ is transmitted.

Using (4.19) and (4.20) we can transform the above into polar coordinates.[4]

$$
\begin{aligned}
P_s &= 1 - \int_{Z_i} \frac{1}{\pi N_o} \exp\{-\frac{1}{N_o}[\rho^2 + E - 2\rho\sqrt{E}\cos(\theta_i - \widehat{\theta})]\}\rho d\rho d\theta \\
&= 1 - \int_{Z_i} p(\rho, \widehat{\theta}/H_i) d\rho d\theta
\end{aligned}
$$

where

$$
p(\rho, \widehat{\theta}/H_i) = \frac{\rho}{\pi N_o} \exp\{-\frac{1}{N_o}[\rho^2 + E - 2\rho\sqrt{E}\cos(\theta_i - \widehat{\theta})]\}
$$

is the joint probability density of ρ and θ. We define $\varphi = \widehat{\theta} - \theta_i$, which represents the phase deviation of the received signal from the transmitted one. Integrating both sides of the above with respect to ρ yields the PDF of $\varphi \in [-\pi, \pi]$ (see Appendix 4A for derivation).

$$
\begin{aligned}
p(\varphi/H_i) &= \frac{e^{-E/N_o}}{2\pi} \left\{ 1 + \sqrt{\frac{\pi E}{N_o}} (\cos\varphi) e^{(E/N_o)\cos^2\varphi} \right. \\
&\quad \left. \cdot \left[1 + \operatorname{erf}\left(\sqrt{\frac{E}{N_o}} \cos\varphi \right) \right] \right\} \\
&= p(\varphi)
\end{aligned}
$$

where

$$
\operatorname{erf}(x) \triangleq \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt
$$

is the *error function*. Note the distribution of φ is independent of index i. This is intuitively correct since φ is the phase deviation, not the absolute phase.

The symbol error probability is the probability that $\widehat{\theta}$ is outside the decision region, or the deviation φ is greater than π/M in absolute value.

$$
P_s = 1 - \int_{-\pi/M}^{\pi/M} p(\varphi) d\varphi \tag{4.21}
$$

When $M = 2$ (BPSK) and $M = 4$ (QPSK) this integration results in the formulas given by (4.6) and (4.37). For $M > 4$, this expression cannot be evaluated in a closed form, the symbol error probability can be obtained by numerically integrating (4.21).

---

[4] Note that $dr_1 dr_2 = \rho d\rho d\widehat{\theta}$.

Another form of P_s is given in [2, p. 209]. The derivation is very complicated and is omitted here. The result is

$$P_s = \frac{M-1}{M} - \frac{1}{2}\operatorname{erf}\left[\sqrt{\frac{E}{N_o}}\sin\frac{\pi}{M}\right] - \frac{1}{\sqrt{\pi}}\int_0^{\sqrt{E/N_o}\sin\pi/M} e^{-y^2}\operatorname{erf}(y\cot\frac{\pi}{M})dy \tag{4.22}$$

This again can only be numerically evaluated for $M > 4$.

Figure 4.13 shows P_s curves for $M = 2, 4, 8, 16$, and 32 given by the exact expression (4.22). Beyond $M = 4$, doubling the number of phases, or increasing one bit in the n-tuples represented by the phases, requires a substantial increase in SNR. For example, at $P_s = 10^{-5}$, the SNR difference between $M = 4$ and $M = 8$ is approximately 4 dB, the difference between $M = 8$ and $M = 16$ is approximately 5 dB. For large values of M, doubling the number of phases requires an SNR increase of 6 dB to maintain the same performance.

For $E/N_o \gg 1$, we can derive an approximation of the P_s expression. First we can use the approximation[5]

$$\operatorname{erf}(x) \approx 1 - \frac{e^{-x^2}}{\sqrt{\pi}x}, \quad x \gg 1$$

to obtain the approximation of the PDF of the phase deviation

$$p(\varphi/H_i) \approx \sqrt{\frac{E}{\pi N_o}}(\cos\varphi)e^{-(E/N_o)\sin^2\varphi} \tag{4.23}$$

Finally substituting (4.23) into (4.21) we arrive at the result

$$\begin{aligned} P_s &\approx \operatorname{erf}c\left(\sqrt{\frac{E}{N_o}}\sin\frac{\pi}{M}\right) \\ &= 2Q\left(\sqrt{\frac{2E}{N_o}}\sin\frac{\pi}{M}\right), \text{ (coherent MPSK)} \end{aligned} \tag{4.24}$$

where

$$\operatorname{erf}c(x) = 1 - \operatorname{erf}(x) = 2Q(\sqrt{2}x)$$

[5] In fact $1 - \frac{e^{-x^2}}{\sqrt{\pi}x}$ is a lower bound of $\operatorname{erf}(x)$, however they are extremely close for $x >> 1$.

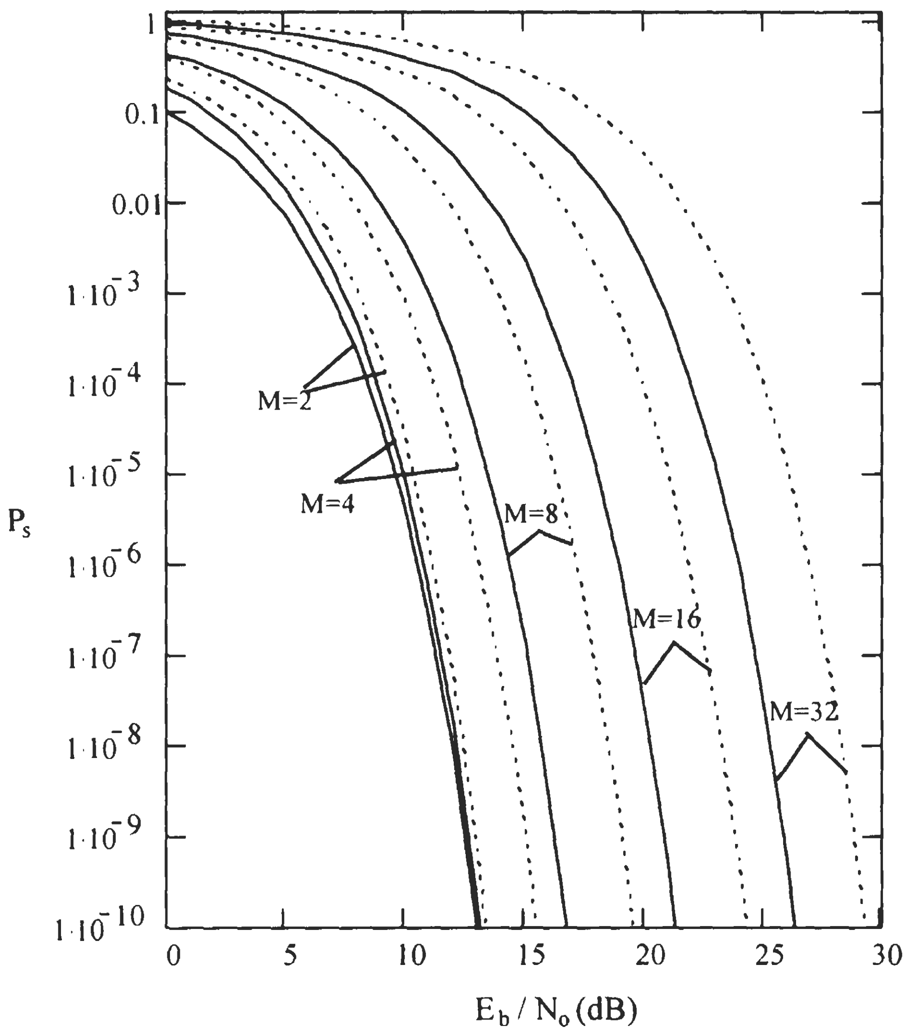


Figure 4.13 P_s of MPSK (solid lines) and DMPSK (dotted lines).

is the *complementary error function.* Note that only high signal-to-noise ratio assumption is needed for the approximation. Therefore (4.24) is good for any values of M, even though it is not needed for $M = 2$ and 4 since precise formulas are available.

Expression (4.24) can be derived geometrically. Consider Figure 4.14. Due to symmetry of the signal constellation, P_s is equal to the error probability of detecting $\mathbf{s}_1$, which is the probability that the received signal vector $\mathbf{r}$ does not fall in the decision region Z_1. This is bounded below and above as follows

$$\Pr(\mathbf{r} \in A_1) \le P_s < \Pr(\mathbf{r} \in A_1) + \Pr(\mathbf{r} \in A_2) = 2\Pr(\mathbf{r} \in A_2)$$

where the equal sign on the left part of the inequality accounts for the case of $M = 2$. The distance from $\mathbf{s}_1$ to the nearest signal is

$$d_{12} = d_{18} = 2\sqrt{E}\sin\frac{\pi}{M}$$

Since white Gaussian noise is identically distributed along any set of orthogonal axes [3, Chapter 3], we may temporarily choose the first axis in such a set as one that passes through the points $\mathbf{s}_1$ and $\mathbf{s}_2$, then for high SNR

$$\begin{aligned}\Pr(\mathbf{r} \in A_1) &= \Pr(\mathbf{r} \in A_2) \approx \Pr(\mathbf{s}_2/H_1) \\ &= \int_{-\infty}^{-d_{12}/2} \frac{1}{\sqrt{\pi N_o}} \exp\{-\frac{x^2}{N_o}\} dx = Q\left(\sqrt{\frac{2E}{N_o}}\sin\frac{\pi}{M}\right)\end{aligned}$$

Thus

$$Q\left(\sqrt{\frac{2E}{N_o}}\sin\frac{\pi}{M}\right) \le P_s \lesssim 2Q\left(\sqrt{\frac{2E}{N_o}}\sin\frac{\pi}{M}\right)$$

Since the lower and upper bounds differ only by a factor of two, which translates into a very small difference in term of SNR, these bounds are very tight.

The bit error rate can be related to the symbol error rate by

$$P_b \approx \frac{P_s}{\log_2 M} \tag{4.25}$$

for Gray coded MPSK signals since most likely the erroneous symbols are the adjacent signals which only differ by one bit.

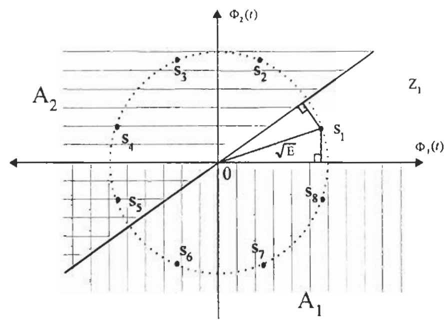


Figure 4.14 Decision regions for bounding P_s of MPSK signals.

4.4 PSD OF MPSK

The PSD of MPSK is similar to that of BPSK except that the spectral is narrower on a frequency scale normalized to the bit rate. As for all carrier modulated signals, it suffices to find the PSD of the complex envelope (Appendix A).

Substituting (4.16) and (4.17) into (4.15), we can write the MPSK signal as

$$s(t) = \mathrm{Re}\left\{\left[\sum_{k=-\infty}^{\infty} A\exp(j\theta_k)p(t-kT)\right]\exp(j2\pi f_c t)\right\}$$

Thus the complex envelope of MPSK is

$$\begin{aligned}\tilde{s}(t) &= \sum_{k=-\infty}^{\infty} A\exp(j\theta_k)p(t-kT) \\ &= \sum_{k=-\infty}^{\infty} A\cos\theta_k p(t-kT) + j\sum_{k=-\infty}^{\infty} A\sin\theta_k p(t-kT)\end{aligned}$$

where

$$\cos\theta_k \in \left(\cos\frac{(2i-1)\pi}{M}, i = 1, 2...M\right)$$

is a random variable which has $M/2$ different values with equal probabilities ($\frac{2}{M}$). Refer to the example of 8-PSK in Figure 4.9. We can see that $\cos\theta_k = \cos\frac{\pi}{8}$ or $\cos\frac{3\pi}{8}$ or $\cos\frac{5\pi}{8}$ or $\cos\frac{7\pi}{8}$, which is $\cos\theta_k = 0.924$ or 0.383 or -0.383 or -0.924. These values are symmetrical about zero. Thus the mean value is zero. The mean square value is

$$\sigma^2 = \sum_{i=1}^{M/2} \frac{2}{M}\cos^2\frac{(2i-1)\pi}{M} = \frac{1}{2}$$

Note that the mean square value is always $\frac{1}{2}$ for $M = 2^n$, $n > 1$. The distribution of $\sin\theta_k$ is the same.

Thus the complex envelope can be written as

$$\tilde{s}(t) = \sum_{k=-\infty}^{\infty} x_k p(t-kT) + j\sum_{k=-\infty}^{\infty} y_k p(t-kT)$$

where $\{x_k = \cos\theta_k\}$ and $\{y_k = \sin\theta_k\}$ are independent, identically distributed random sequences with zero means and a mean square value of $1/2$. The PSD of this type of complex envelope has been derived in Appendix A. The result (A.21) can be directly used here.

Since $\sigma_x^2 = \sigma_y^2 = \sigma^2 = 1/2$ and

$$|P(f)| = \left|AT\frac{\sin\pi fT}{\pi fT}\right|$$

then from (A.21) we have

$$\begin{aligned}\Psi_{\tilde{s}}(f) &= 2\sigma^2 A^2 T\left(\frac{\sin\pi fT}{\pi fT}\right)^2 \\ &= A^2 T\left(\frac{\sin\pi fT}{\pi fT}\right)^2 \\ &= A^2 nT_b\left(\frac{\sin\pi fnT_b}{\pi fnT_b}\right)^2, \text{ (MPSK)} \qquad (4.26)\end{aligned}$$

where $n = \log_2 M$. This is exactly the same as that of BPSK in terms of symbol rate. However, in terms of bit rate the PSD of MPSK is n-times narrower than the

BPSK. Figure 4.15 is the PSDs ($A = \sqrt{2}$ and $T_b = 1$ for unit bit energy: $E_b = 1$) for different values of M where the frequency axis is normalized to the bit rate (fT_b).

Since the passband minimum (Nyquist) bandwidth required to transmit the symbols is $1/T$, the maximum bandwidth efficiency is

$$\frac{R_b}{B_{\min}} = \frac{(\log_2 M)/T}{1/T} = \log_2 M$$

4.5 DIFFERENTIAL MPSK

In Section 4.2 we discussed DBPSK, which is in fact a special case of differential MPSK (DMPSK). The term DMPSK refers to "differentially encoded and differentially coherently demodulated MPSK." The differentially coherent demodulation is in fact noncoherent in the sense that phase coherent reference signals are not required. It is used to overcome the adversary effect of the random phase in the received signal.

Differentially encoded MPSK can also be coherently demodulated (denoted as DEMPSK). In this case, the purpose of differential encoding is to eliminate phase ambiguity in the carrier recovery process. This is not usually meant by the term DMPSK.

In both cases, the modulation processes are the same. In other words, the transmitted MPSK signals are the same. Only demodulations are different.

In the modulator the information bits are first differentially encoded. Then the encoded bits are used to modulate the carrier. In a DEMPSK signal stream, information is carried by the phase difference $\Delta\theta_i$ between two consecutive symbols. There are M different values of $\Delta\theta_i$, each represents an n-tuple ($n = \log_2 M$) of information bits.

For $M = 2$ and 4, encoding, modulation, and demodulation are simple, as we have seen in Section 4.2 for DBPSK and will see shortly for differentially encoded QPSK.

In light of the modern digital technology, DEMPSK signals can be generated by digital frequency synthesis technique. A phase change from one symbol to the next is simply controlled by the n-tuple which is represented by the phase change. This technique is particularly suitable for large values of M.

In DMPSK scheme, the DEMPSK signal is demodulated by a differentially coherent (or optimum noncoherent) demodulator as shown in Figure 4.16.

The derivation of the demodulator is similar to that of binary DPSK. In DEMPSK a message m_i of $n = \log_2 M$ bits is represented by the phase difference of two consecutive symbols. In other words, m_i is represented by a symbol with two symbol

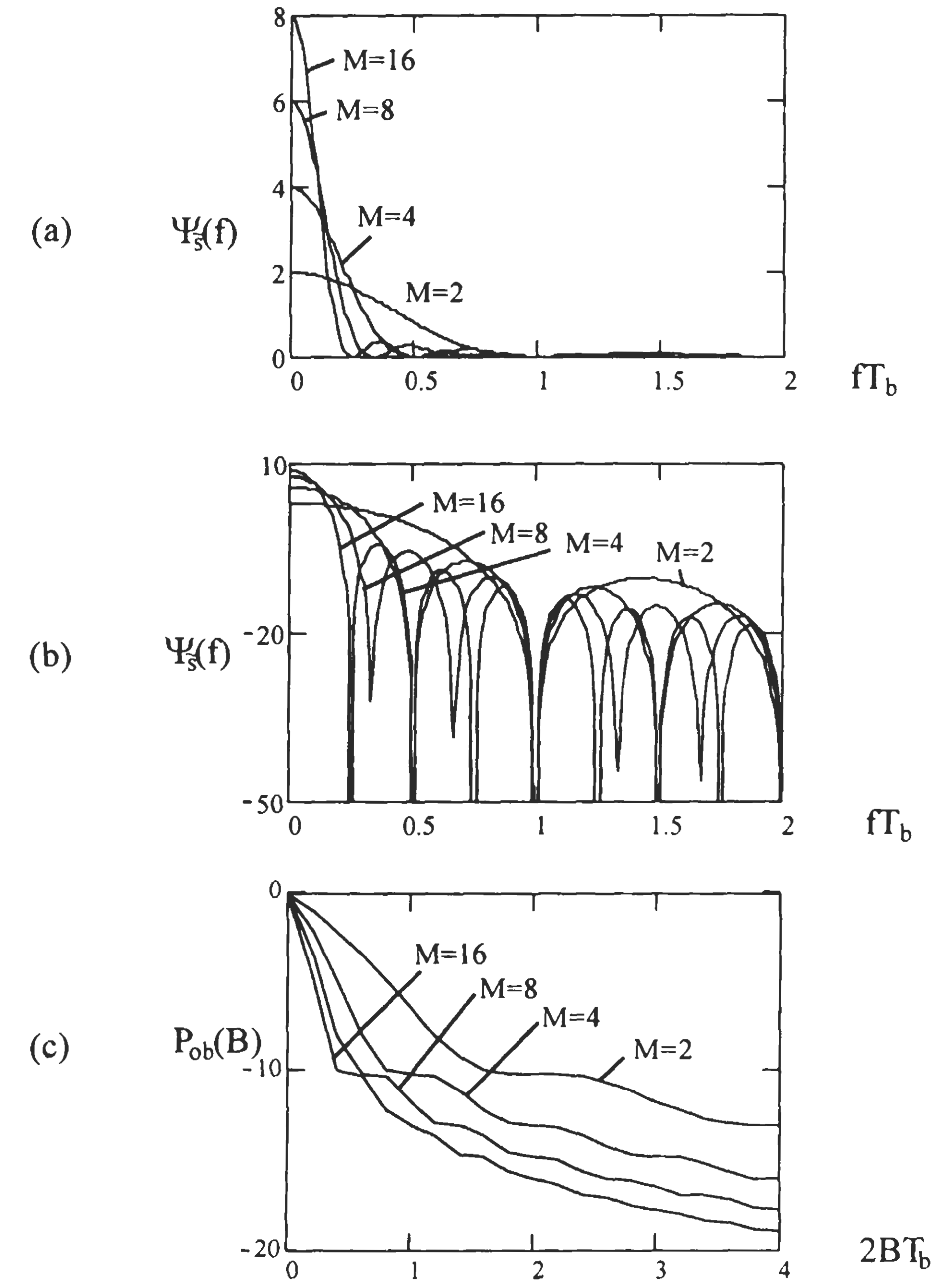


Figure 4.15 PSDs of MPSK: (a) linear, (b) logarithmic, (c) out-of-band power.

periods defined as

$$\xi_i(t) = \begin{cases} A\cos\left[2\pi f_c t + \Phi_0\right], & 0 \le t \le T \\ A\cos\left[2\pi f_c t + \Phi_0 + \Delta\theta_i\right], & T \le t \le 2T \end{cases}$$

where$\Delta\,\theta_i = \frac{2(i-1)\pi}{M}, i = 1, 2, \ldots M$. The received signal $r(t)$ has an unknown phase θ introduced by the channel and is corrupted by AWGN. Consider the first two symbol durations $[0 \le t \le 2T]$, from (B.55), the sufficient statistic for detecting $\xi_i(t)$ is

$$l_i^2 = \left(\int_0^{2T} r(t)\xi_i(t)dt\right)^2 + \left(\int_0^{2T} r(t)\xi_i(t, \frac{\pi}{2})dt\right)^2 \tag{4.27}$$

the first integral is

$$\begin{aligned}
&\int_0^{2T} r(t)\xi_i(t)dt \\
= &\int_0^T r(t)A\cos(2\pi f_c t + \Phi_0)dt \\
&+ \int_T^{2T} r(t)A\cos(2\pi f_c t + \Phi_0 + \Delta\theta_i)dt \\
= &\int_0^T r(t)A\left[\cos 2\pi f_c t\cos\Phi_0 - \sin 2\pi f_c t\sin\Phi_0\right]dt \\
&+ \int_T^{2T} r(t)A\left[\cos 2\pi f_c t\cos(\Phi_0 + \Delta\theta_i) - \sin 2\pi f_c t\sin(\Phi_0 + \Delta\theta_i)\right]dt \\
= &\; w_0\cos\Phi_0 + z_0\sin\Phi_0 + w_1\cos(\Phi_0 + \Delta\theta_i) + z_1\sin(\Phi_0 + \Delta\theta_i)
\end{aligned}$$

where

$$\begin{aligned}
w_0 &\triangleq \int_0^T r(t)A\cos 2\pi f_c t dt \\
z_0 &\triangleq -\int_0^T r(t)A\sin 2\pi f_c t dt \\
w_1 &\triangleq \int_T^{2T} r(t)A\cos 2\pi f_c t dt \\
z_1 &\triangleq -\int_T^{2T} r(t)A\sin 2\pi f_c t dt
\end{aligned}$$

and the second integral is

$$\begin{aligned}
&\int_0^{2T} r(t)\xi_i(t, \frac{\pi}{2})dt \\
= &\int_0^T r(t)A\sin(2\pi f_c t + \Phi_0)dt + \int_T^{2T} r(t)A\sin(2\pi f_c t + \Phi_0 + \Delta\theta_i)dt \\
= &-z_0\cos\Phi_0 + w_0\sin\Phi_0 - z_1\cos(\Phi_0 + \Delta\theta_i) + w_1\sin(\Phi_0 + \Delta\theta_i)
\end{aligned}$$

Then substituting these two integrals into (4.27), expanding the squares, discarding squared terms since they are independent of transmitted signals, dropping a factor of two, we have the following new sufficient statistic

$$L_i = (w_1 w_0 + z_1 z_0)\cos\Delta\theta_i + (z_1 w_0 - w_1 z_0)\sin\Delta\theta_i$$

For the kth symbol duration this is

$$\begin{aligned}
L_i &= \underbrace{(w_k w_{k-1} + z_k z_{k-1})}_{x_k}\cos\theta_i + \underbrace{(z_k w_{k-1} - w_k z_{k-1})}_{y_k}\sin\theta_i \\
&= x_k\cos\Delta\theta_i + y_k\sin\Delta\theta_i
\end{aligned} \tag{4.28}$$

The decision rule is to choose the largest. Or we can write (4.28) as

$$\begin{aligned}
L_i &= A\cos\Delta\widehat{\theta}_k\cos\Delta\theta_i + A\sin\Delta\widehat{\theta}_k\sin\Delta\theta_i \\
&= A\cos(\Delta\theta_i - \Delta\widehat{\theta}_k)
\end{aligned}$$

where

$$\Delta\widehat{\theta}_k = \tan^{-1}\frac{y_k}{x_k}$$

Thus the decision rule is to choose the smallest $|\Delta\theta_i - \Delta\widehat{\theta}_k|$. Figure 4.16 implements this rule. As we stated in the binary DPSK case, the local oscillator output must have the same frequency, but not necessarily the same phase, as the received signal. The amplitude of the reference signals can be any value, which is unit in the figure, since the effect of the amplitude is cancelled when computing $\Delta\widehat{\theta}_k$.

The symbol error probability is given by [2]

$$P_s = \frac{\sin\frac{\pi}{M}}{2\pi}\int_{-\pi/2}^{\pi/2}\frac{\exp\{-\frac{E}{N_o}[1 - \cos\frac{\pi}{M}\cos x]\}}{1 - \cos\frac{\pi}{M}\cos x}dx \tag{4.29}$$

which can be evaluated in a closed form for $M = 2$ (see (4.10)). For other values of M, it can only be numerically evaluated. Many approximate expressions have been

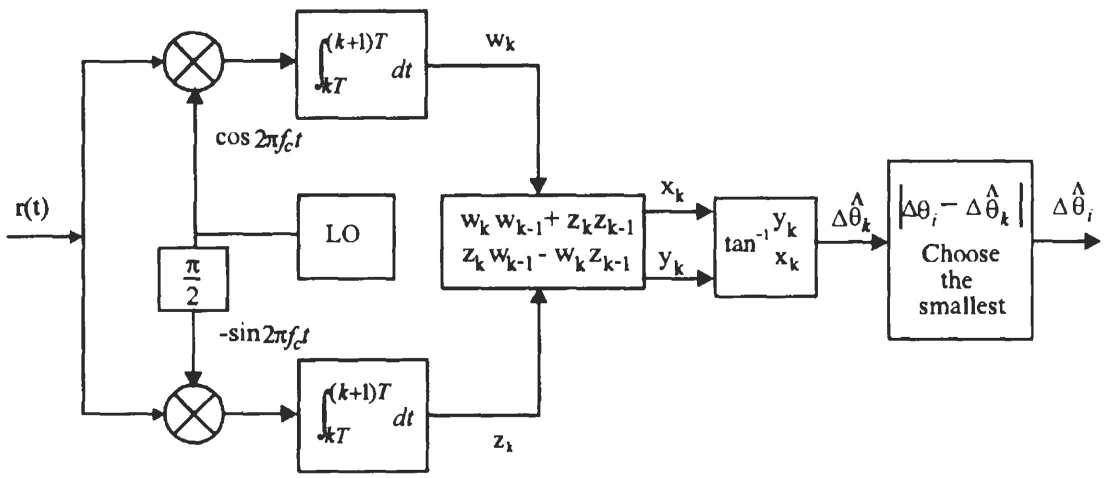


Figure 4.16 Differentially coherent demodulator for differentially encoded MPSK signals.

found [2,4], a simple one is

$$P_s \approx 2Q\left(\sqrt{\frac{2E}{N_o}}\sin\frac{\pi}{\sqrt{2}M}\right), \text{ (optimum DMPSK)} \tag{4.30}$$

for large SNR. The exact curves as obtained from (4.29) are given in Figure 4.13 together with those of coherent MPSK. Compared with coherent MPSK, asymptotically the DMPSK requires 3 dB more SNR to achieve the same error performance. This also can be quite easily seen by comparing the arguments of the Q-function in (4.30) and (4.24), using $sin(x) \approx x$ for small x.

For the purpose of phase ambiguity elimination, the DEMPSK signal is coherently demodulated. The optimum demodulator is shown in Figure 4.17 which is similar to Figure 4.11, the demodulator for coherent MPSK, except that a differential decoder is attached as a final stage. This is intuitively convincing since at carrier frequency the DEMPSK signal is the same as MPSK signal, thus the correlator part is the same as that of coherent MPSK. The additional differential decoder recovers the differential phase $\Delta\theta_i$ from phases of two consecutive symbols. The $\Delta\theta_i$ then is mapped back to the corresponding n-tuple of bits. Rigorous derivation of this optimum demodulator and its equivalent forms can be found in [2,4].

The symbol error probability of coherently demodulated DEMPSK is given by [2,4]

$$P_s = 2P_{s-MPSK}\left[1 - \frac{1}{2}P_{s-MPSK} - \frac{1}{2}\frac{\sum_{i=1}^{M-1} P_i(C)}{P_{s-MPSK}}\right], \text{ (DEMPSK)} \tag{4.31}$$

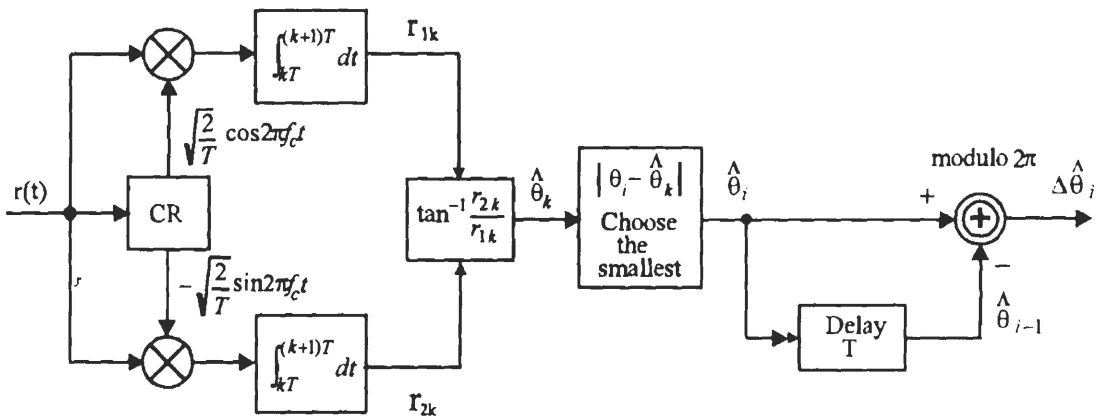


Figure 4.17 Coherent demodulation of differentially encoded MPSK signals.

where P_{s-MPSK} is the symbol error probability for MPSK without differential encoding and is given by (4.21), and

$$P_i(C) = \left[\int_{(2i-1)\pi/M}^{(2i+1)\pi/M}\int_0^\infty \frac{r}{\pi}\exp\left\{-\left[r^2 - 2r\sqrt{\frac{E}{N_o}}\cos\Theta + \frac{E}{N_o}\right]\right\}drd\Theta\right]^2$$

For $M = 2$, we have coherent detection of differentially encoded BPSK for which (4.31) reduces to (4.12). For $M = 4$, it reduces to

$$\begin{aligned} P_s &= 4Q\left(\sqrt{\frac{E}{N_o}}\right) - 8\left[Q\left(\sqrt{\frac{E}{N_o}}\right)\right]^2 \\ &\quad +8\left[Q\left(\sqrt{\frac{E}{N_o}}\right)\right]^3 - 4\left[Q\left(\sqrt{\frac{E}{N_o}}\right)\right]^4 \end{aligned} \tag{4.32}$$

For large SNR, the second, third, and fourth terms can be ignored. Thus the above is just about two times that of coherent QPSK without differential encoding. In fact for any value of M when the SNR is large, the terms in the bracket of (4.31) is close to one, thus the P_s of the coherently demodulated DEMPSK is about two times that of coherent MPSK without differential encoding. This translates to 0.5 dB or less degradation in SNR. This is the price paid for removing the phase ambiguity.

A DEMPSK signal's PSD would be the same as its nonencoded counterpart if the encoding process does not change the statistic characteristic of the baseband data, since the final signal from the modulator is just MPSK signal. We always assume that the original data have an equally likely distribution. This results in that the distribution of $\Delta\theta_i$ is equally likely too. In turn the absolute phases of the DEMPSK signals are also equally likely. This satisfies the condition for deriving (4.26). Thus the PSD of DEMPSK is the same as that of MPSK given in (4.26) for an equally likely original data sequence.

As we have proved in Chapter 2 and mentioned in Section 4.2, that differential encoding in DEBPSK always produces an equally likely data sequence asymptotically regardless of the distribution of the original data. This leads to a PSD given by (4.8) for DEBPSK even if the original data is not evenly distributed.

4.6 QUADRATURE PSK

Among all MPSK schemes, QPSK is the most often used scheme since it does not suffer from BER degradation while the bandwidth efficiency is increased. Other MPSK schemes increase bandwidth efficiency at the expenses of BER performance. In this section we will study QPSK in great detail.

Since QPSK is a special case of MPSK, its signals are defined as

$$s_i(t) = A\cos(2\pi f_c t + \theta_i), \qquad 0 \le t \le T, \quad i = 1, 2, 3, 4 \tag{4.33}$$

where

$$\theta_i = \frac{(2i-1)\pi}{4}$$

The initial signal phases are $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$. The carrier frequency is chosen as integer multiple of the symbol rate, therefore in any symbol interval $[kT, (k+1)T]$, the signal initial phase is also one of the four phases.

The above expression can be written as

$$\begin{aligned} s_i(t) &= A\cos\theta_i \cos 2\pi f_c t - A\sin\theta_i \sin 2\pi f_c t \\ &= s_{i1}\phi_1(t) + s_{i2}\phi_2(t) \end{aligned} \tag{4.34}$$

where $\phi_1(t)$ and $\phi_2(t)$ are defined in (4.2) and (4.3),

$$s_{i1} = \sqrt{E}\cos\theta_i$$

$$s_{i2} = \sqrt{E}\sin\theta_i$$

and

$$\theta_i = \tan^{-1}\frac{s_{i2}}{s_{i1}}$$

Dibit	Phase θ_i	$s_{i1} = \sqrt{E}\cos\theta_i$	$s_{i2} = \sqrt{E}\sin\theta_i$
11	$\pi/4$	$+\sqrt{E/2}$	$+\sqrt{E/2}$
01	$3\pi/4$	$-\sqrt{E/2}$	$+\sqrt{E/2}$
00	$-3\pi/4$	$-\sqrt{E/2}$	$-\sqrt{E/2}$
10	$-\pi/4$	$+\sqrt{E/2}$	$-\sqrt{E/2}$

Table 4.2 QPSK signal coordinates.

where $E = A^2T/2$ is the symbol energy. We observe that this signal is a linear combination of two orthonormal basis functions: $\phi_1(t)$ and $\phi_2(t)$. On a coordinate system of $\phi_1(t)$ and $\phi_2(t)$ we can represent these four signals by four points or vectors: $\mathbf{s}_i = \begin{bmatrix} s_{i1} \\ s_{i2} \end{bmatrix}, i = 1, 2, 3, 4$. The angle of vector $\mathbf{s}_i$ with respect to the horizontal axis is the signal initial phase θ_i. The length of the vectors is $\sqrt{E}$.

The signal constellation is shown in Figure 4.18. In a QPSK system, data bits are divided into groups of two bits, called dibits. There are four possible dibits, 00, 01, 10, and 11. Each of the four QPSK signals is used to represent one of them. The mapping of the dibits to the signals could be arbitrary as long as the mapping is one to one. The signal constellation in Figure 4.18 uses the Gray coding. The coordinates of signal points are tabulated in Table 4.2.

In the table, for convenience of modulator structure, we map logic 1 to $\sqrt{E/2}$ and logic 0 to $-\sqrt{E/2}$. We also map odd-numbered bits to s_{i1} and even-numbered bits to s_{i2}. Thus from (4.34) the QPSK signal on the entire time axis can be written as

$$s(t) = \frac{A}{\sqrt{2}} I(t) \cos 2\pi f_c t - \frac{A}{\sqrt{2}} Q(t) \sin 2\pi f_c t, \quad -\infty < t < \infty \tag{4.35}$$

where $I(t)$ and $Q(t)$ are pulse trains determined by the odd-numbered bits and even-numbered bits, respectively.

$$I(t) = \sum_{k=-\infty}^{\infty} I_k p(t - kT)$$

$$Q(t) = \sum_{k=-\infty}^{\infty} Q_k p(t - kT)$$

where $I_k = \pm 1$ and $Q_k = \pm 1$, the mapping between logic data and I_k or Q_k is $1 \to 1$ and $0 \to -1$. $p(t)$ is a rectangular pulse shaping function defined on $[0, T]$.

The QPSK waveform using the signal assignment in Figure 4.18 is shown in Figure 4.19. Like BPSK, the waveform has a constant envelope and discontinuous

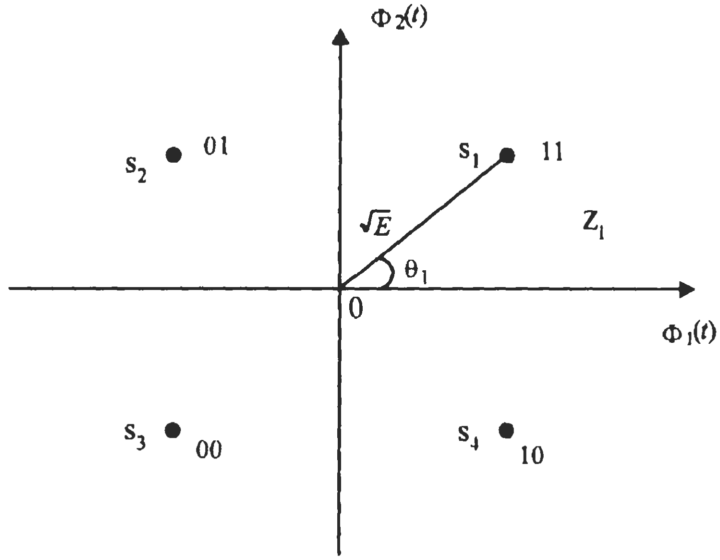


Figure 4.18 QPSK signal constellation.

phases at symbol boundaries. But unlike BPSK, the symbol interval is $2T_b$ instead of T_b. If the transmission rate of the symbols is the same in QPSK and BPSK, it is intuitively clear that QPSK transmits data twice as fast as BPSK does. Also we observe that the distance of adjacent points of the QPSK constellation is shorter than that of the BPSK. Does this cause the demodulator more difficulty, in comparison with BPSK, to distinguish those symbols, therefore symbol error performance is degraded and consequently bit error rate is also degraded ? Surprisingly, it turns out that even though symbol error probability is increased, the bit error probability remains unchanged, as we will see shortly.

The modulator of QPSK is based on (4.35). This leads to the modulator in Figure 4.20(a). The channel with cosine reference is called inphase (I) channel and the channel with sine reference is called quadrature (Q) channel. The data sequence is separated by the serial-to-parallel converter (S/P) to form the odd-numbered-bit sequence for I-channel and the even-numbered-bit sequence for Q-channel. Then logic 1 is converted to a positive pulse and logic 0 is converted to a negative pulse, both have the same amplitude and a duration of T. Next the odd-numbered-bit pulse train is multiplied to $\cos 2\pi f_c t$ and the even-numbered-bit pulse train is multiplied to $\sin 2\pi f_c t$. It is clear that the I-channel and Q-channel signals are BPSK signals with a symbol duration of $2T_b$. Finally a summer adds these two waveforms together to produce the final QPSK signal. (see Figure 4.19 for waveforms at various stages.)

Since QPSK is a special case of MPSK, the demodulator for MPSK (Figure

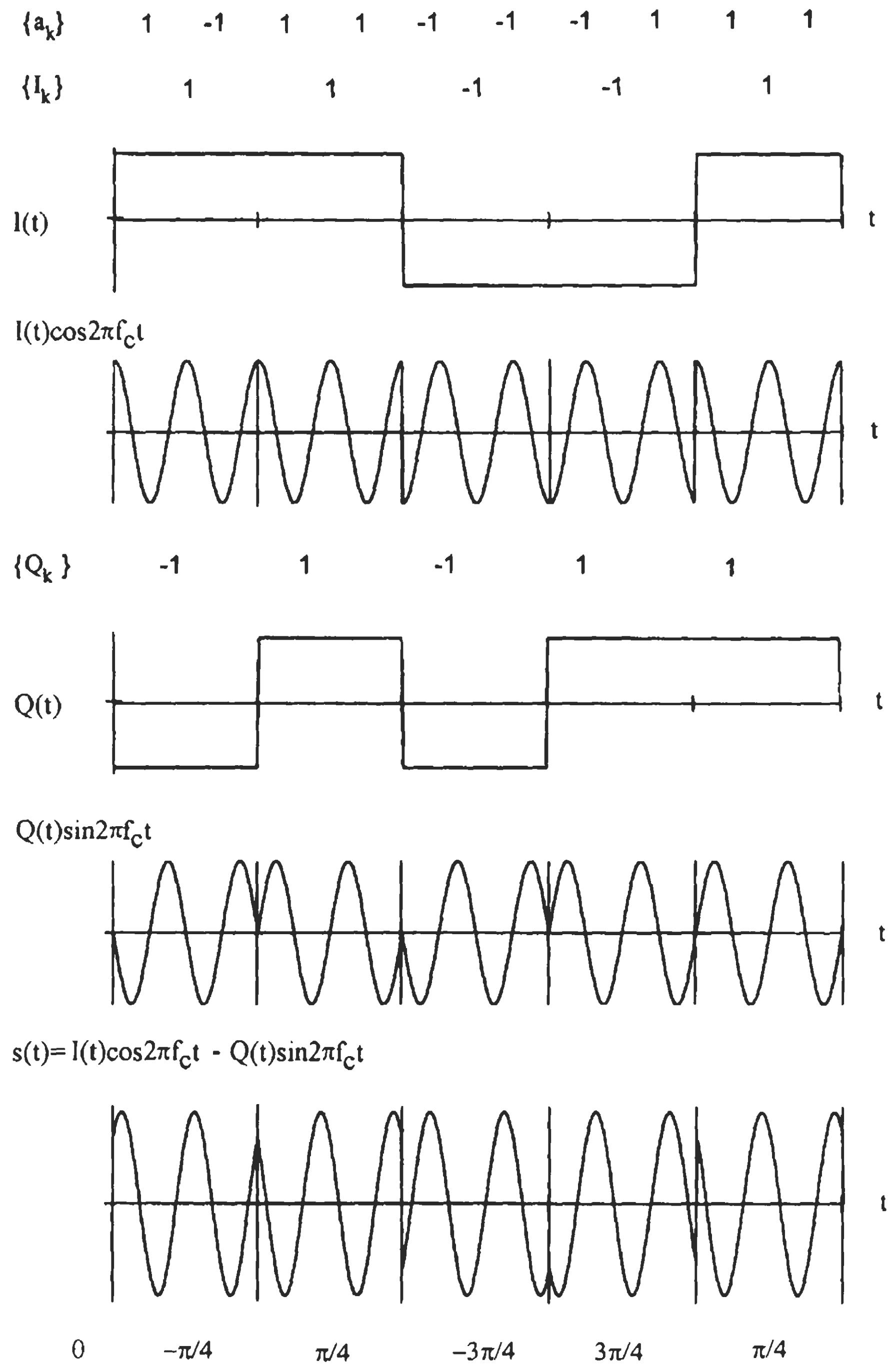
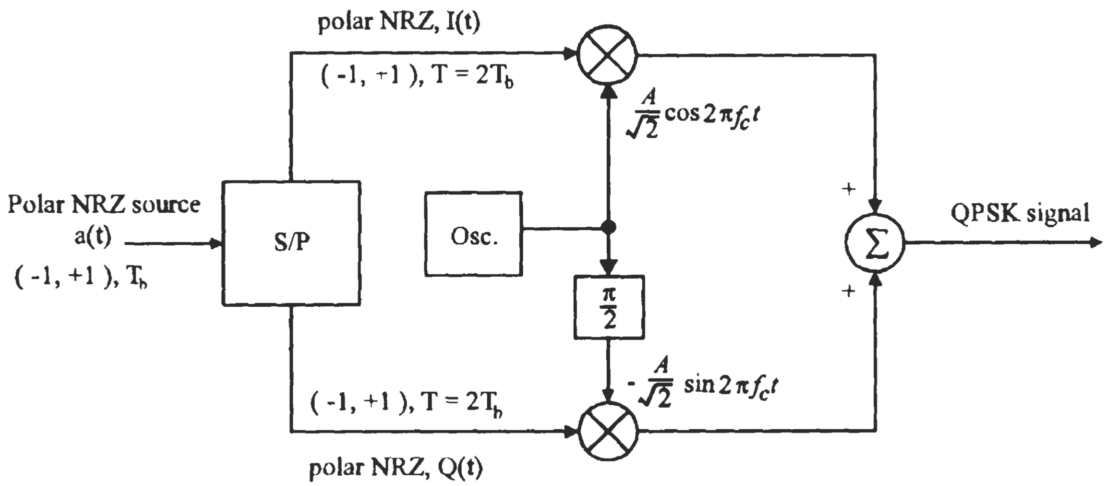
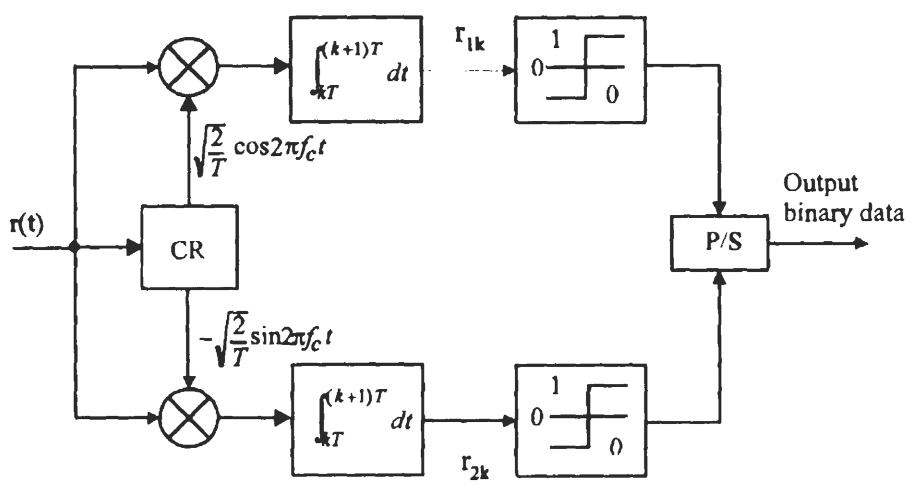


Figure 4.19 QPSK waveforms.



(a)



(b)

Figure 4.20 (a) QPSK modulator, (b) QPSK demodulator.

4.11) is applicable to QPSK. However, due to the special property of the QPSK constellation, a simpler demodulator is possible. It is shown in Figure 4.20(b) which is equivalent to Figure 4.11. I- and Q-channel signals are demodulated separately as two individual BPSK signals. A parallel-to-serial converter (P/S) is used to combine two sequences into a single sequence. This is possible because of the one-to-one correspondence between data bits and I- and Q-channel signals and their orthogonality. For $M > 4$, the optimum receiver can only be the form in Figure 4.11, since the signal in the I-channel or Q-channel does not correspond to a single bit, as we have seen in Section 4.3.

The bit error probability of the optimum demodulators can be derived using the demodulator of Figure 4.20. Since $E\{r_j\}, j = 1, 2$, is either $\sqrt{E/2}$ or $-\sqrt{E/2}$, corresponding to a bit of 1 or 0 (Table 4.2), the detection is a typical binary detection with a threshold of 0. The average bit error probability for each channel is

$$\begin{aligned} P_b &= \Pr(e/1 \text{ is sent}) = \Pr(e/0 \text{ is sent}) \\ &= \int_0^{\infty} \frac{1}{\sqrt{\pi N_o}} \exp\left[-\frac{(R_j + \sqrt{E/2})^2}{N_o}\right] dR_j \\ &= \int_{\sqrt{\frac{E}{N_o}}}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{x^2}{2}\right] dx \\ &= Q\left(\sqrt{\frac{E}{N_o}}\right) = Q\left(\sqrt{\frac{2E_b}{N_o}}\right), \text{ (coherent QPSK)} \end{aligned} \tag{4.36}$$

The final output of the demodulator is just the multiplexed I- and Q-channel outputs. Thus the bit error rate for the final output is the same as that of each channel. A symbol represents two bits from the I- and Q-channels, respectively. A symbol error occurs if any one of them is in error. Therefore the symbol error probability is

$$\begin{aligned} P_s &= 1 - \Pr(\text{ both bits are correct}) \\ &= 1 - (1 - P_b)^2 \\ &= 2P_b - P_b^2 \\ &= 2Q\left(\sqrt{\frac{E}{N_o}}\right) - \left[Q\left(\sqrt{\frac{E}{N_o}}\right)\right]^2 \end{aligned} \tag{4.37}$$

The above symbol error probability expression can also be derived from the general formula in Section 4.3 for MPSK(4.21). Then the bit error probability expression can be derived in another way as follows. First for large SNR, the second term in (4.37) can be ignored. Second, for Gray coding and large SNR, a symbol error most likely causes the symbol being detected as the adjacent symbol which is only one bit different out of two bits. Thus

$$P_b \approx \frac{1}{2} P_s \approx Q\left(\sqrt{\frac{2E_b}{N_o}}\right)$$

This expression is derived by approximations. But it is the same as the one obtained by the accurate derivation. We have made approximations twice. The first is to ignore the second term in (4.37). This increases the estimate of P_b slightly. The second is to ignore the symbol errors caused by choosing the nonadjacent symbols which may cause two bit errors for a symbol error. This decreases the estimate of P_b slightly. The fact that the final estimate is exactly equal to the accurate one shows that these two approximations happen to cancel each other. It is purely a coincidence.

The P_b curve of QPSK is shown in Figure 4.21, which is the same as that of BPSK. The P_s curve of QPSK is shown in Figure 4.13 together with other MPSK schemes.

The PSD of QPSK is similar to that of BPSK except that the spectral is narrower on a frequency scale normalized to the bit rate. From (4.26) we have

$$\Psi_{\tilde{s}}(f) = 2A^2 T_b \left(\frac{\sin 2\pi f T_b}{2\pi f T_b}\right)^2, \text{ (QPSK)} \tag{4.38}$$

Figure 4.22(a, b) are the PSD curves of the QPSK. The null-to-null bandwidth $B_{null} = 1/T_b = R_b$. Figure 4.22(c) is the out-of-band power curve from which we can estimate that $B_{90\%} \approx 0.75R_b$. We also calculated that $B_{99\%} \approx 8R_b$.

4.7 DIFFERENTIAL QPSK

Now we study an important special case of DEMPSK, the DEQPSK. In DEQPSK information dibits are represented by the phase differences $\Delta\theta_i$ from symbol to symbol. There are different phase assignments between $\Delta\theta_i$ and logic dibits. A possible phase assignment is listed in Table 4.3. Our discussion in this section is based on this phase assignment choice (later when we study $\pi/4-$QPSK, the phase assignment is different). An example for this choice is shown in Table 4.4.

The coding rules are as follows [4].

$$\begin{aligned} u_k &= \overline{(I_k \oplus Q_k)}(I_k \oplus u_{k-1}) + (I_k \oplus Q_k)(Q_k \oplus v_{k-1}) \\ v_k &= \overline{(I_k \oplus Q_k)}(Q_k \oplus v_{k-1}) + (I_k \oplus Q_k)(I_k \oplus u_{k-1}) \end{aligned} \tag{4.39}$$

where $\oplus$ denotes exclusive OR operation. $I_k \in (0, 1)$ and $Q_k \in (0, 1)$ are odd-numbered and even-numbered original information bits, respectively; $u_k \in (0, 1)$

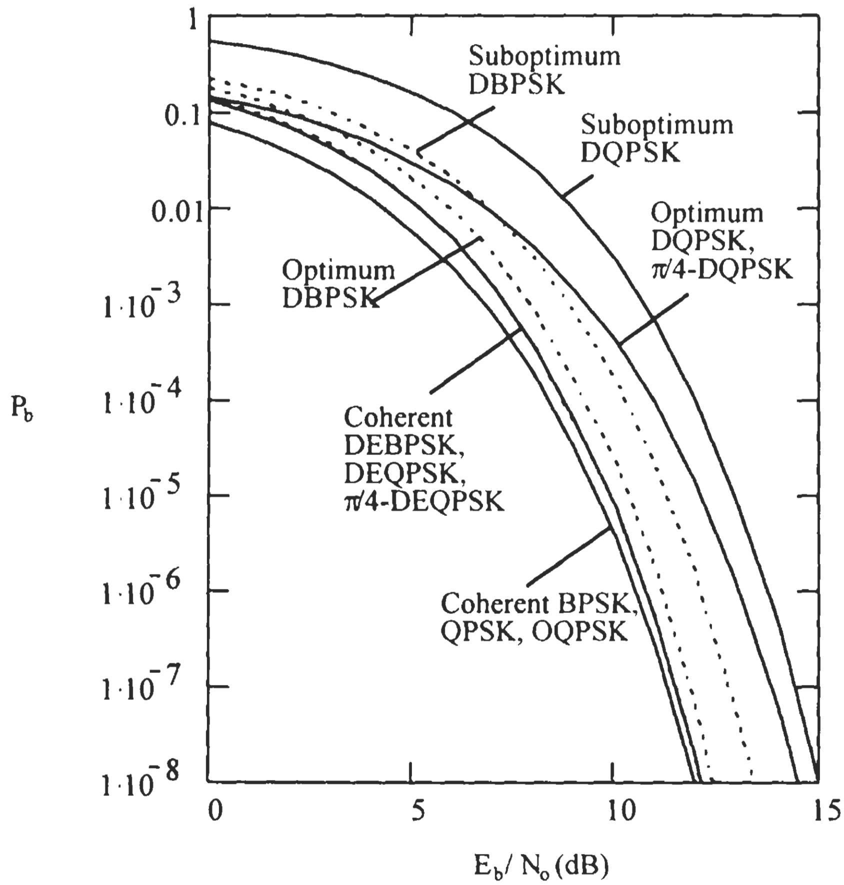


Figure 4.21 P_b of QPSK and DQPSK.

and $v_k \in (0, 1)$ are coded I-channel and Q-channel bits, respectively. Pairs (I_k, Q_k) and (u_{k-1}, v_{k-1}) are used to produce pair (u_k, v_k) which is used to control the absolute phase of the carrier. The resultant signal is a QPSK signal as shown in Figure 4.18 for (u_k, v_k), but it is a DEQPSK signal for (I_k, Q_k). Therefore the modulator is basically the same as the QPSK modulator (Figure 4.20) except that two differential encoders must be included in each channel before the carrier multiplier. The modulator is shown in Figure 4.23.

When DEQPSK is differentially coherently demodulated, the scheme is DQPSK. The optimum DQPSK demodulator can be derived from Figure 4.16 as a special case of $M = 4$. The symbol error probability is given by (4.29) or (4.30). The bit error

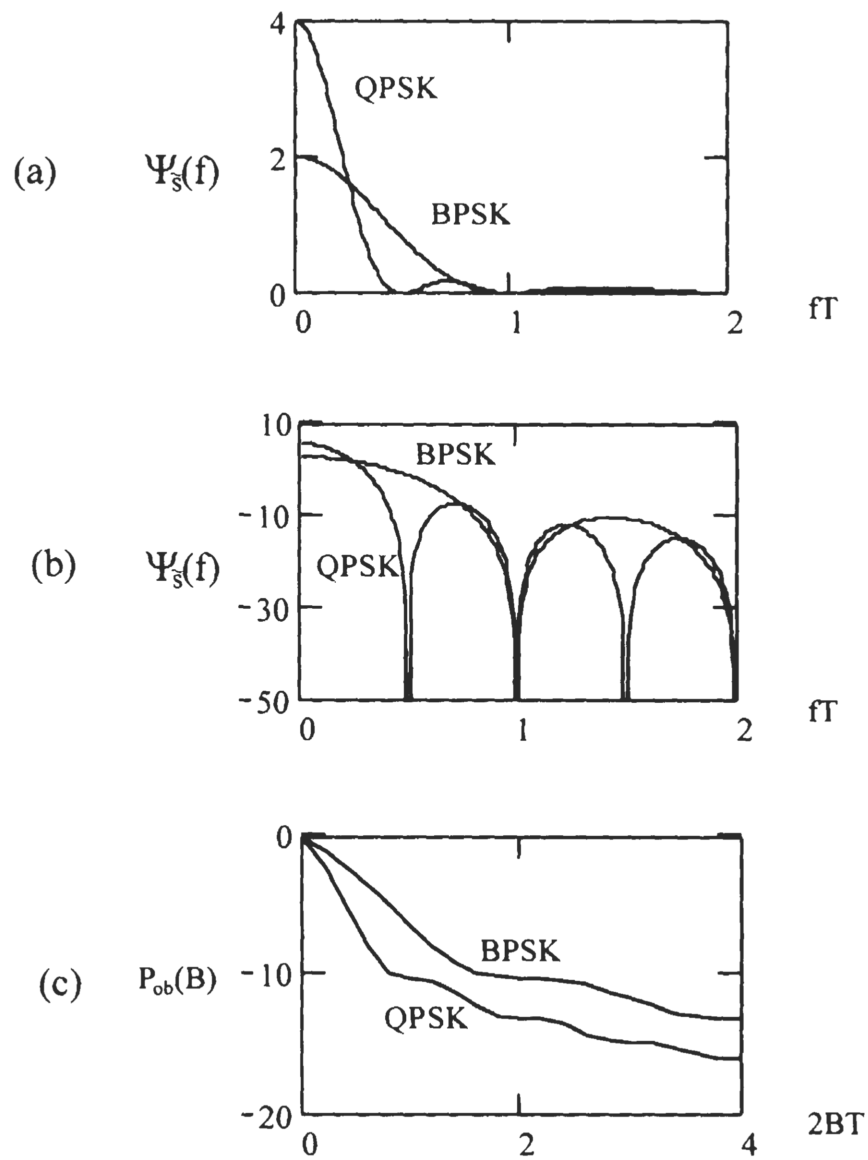


Figure 4.22 PSD of QPSK in comparison with BPSK: (a) linear, (b) logarithmic, (c) out-of-band power.

Dibit	$\Delta\theta_i$	$\cos\Delta\theta_i$	$\sin\Delta\theta_i$
00	0	1	0
01	$\pi/2$	0	1
10	$-\pi/2$	0	-1
11	π	-1	0

Table 4.3 DEQPSK signal phase assignment.

Modulation		ref.								
Information sequence	I_k		1	0	1	0	1	1	0	1
	Q_k		0	1	0	1	1	0	0	1
Encoded sequence	u_k	1	1	1	1	1	0	0	0	1
	v_k	1	0	1	0	1	0	1	1	0
Transmitted absolute phases		$\frac{\pi}{4}$	$\frac{7\pi}{4}$	$\frac{\pi}{4}$	$\frac{7\pi}{4}$	$\frac{\pi}{4}$	$\frac{5\pi}{4}$	$\frac{3\pi}{4}$	$\frac{3\pi}{4}$	$\frac{7\pi}{4}$

Table 4.4 Differential coding for DEQPSK.

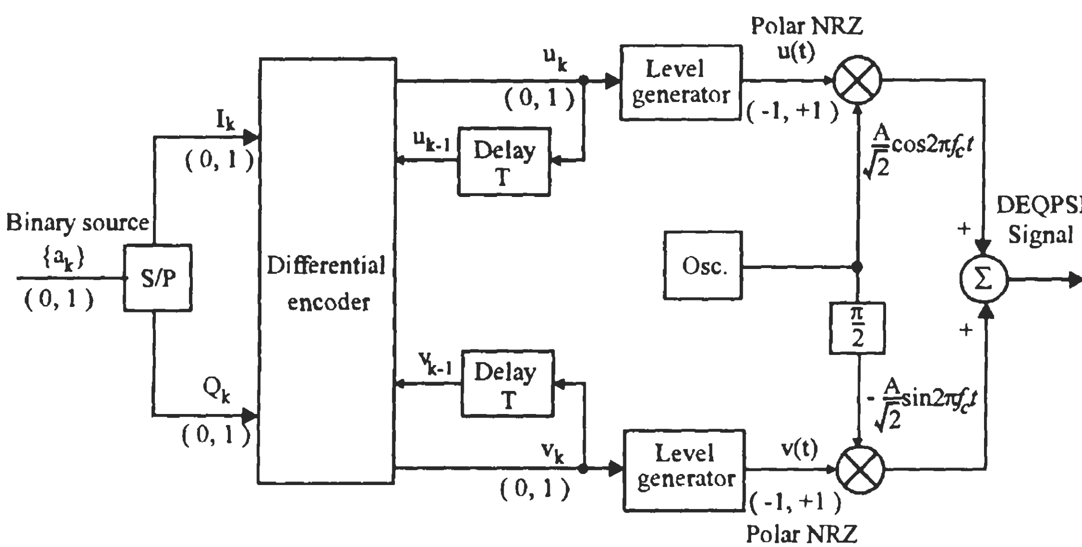


Figure 4.23 DEQPSK modulator.

probability can be approximately calculated using (4.25). Thus

$$P_b \approx Q\left(\sqrt{\frac{4E_b}{N_o}}\sin\frac{\pi}{4\sqrt{2}}\right), \text{ (Optimum DQPSK)} \tag{4.40}$$

which is plotted in Figure 4.21. It is about 2 to 3 dB inferior to coherent QPSK. Alternately, bit error probability of DQPSK can be evaluated using an expression given by [5] as

$$P_b = e^{-2\gamma_b}\sum_{k=0}^{\infty}(\sqrt{2}-1)^k I_k(\sqrt{2}\gamma_b) - \frac{1}{2}I_0(\sqrt{2}\gamma_b)e^{-2\gamma_b} \tag{4.41}$$

where $\gamma_b = E_b/N_o$ and $I_\alpha(x)$ is the αth order modified Bessel function of the first kind which may be represented by the infinite series

$$I_\alpha(x) \triangleq \sum_{k=0}^{\infty}\frac{(x/2)^{\alpha+2k}}{k!\Gamma(\alpha+k+1)}, \quad x \geq 0$$

and the gamma function is defined as

$$\Gamma(p) \triangleq \int_0^\infty t^{p-1}e^{-t}dt, \quad p > 0$$

Like in binary DPSK case, a suboptimum demodulator using previous symbols as references is shown in Figure 4.24 where the integrator can be replaced by a low-pass filter [6]. The front-end bandpass filter reduces noise power but preserves the phase of the signal. In the absence of noise, the I-channel integrator output is

$$\begin{aligned}
&\int_{kT}^{(k+1)T} A^2\cos(2\pi f_c t + \theta_k)\cos(2\pi f_c t + \theta_{k-1})dt \\
&= \frac{1}{2}\int_{kT}^{(k+1)T} A^2[\cos(4\pi f_c t + \theta_k + \theta_{k-1}) + \cos(\theta_k - \theta_{k-1})dt \\
&= \frac{1}{2}A^2 T\cos\Delta\theta_k
\end{aligned}$$

Similarly the Q-channel integrator output is $\frac{1}{2}A^2T\sin\Delta\theta_k$. The arctangent operation extracts the $\Delta\widehat{\theta}_k$ (estimate of $\Delta\theta_k$ with the presence of noise) and a comparator compares it to the four $\Delta\theta_i$ and chooses the closest. The dibit is then recovered from the detected $\Delta\theta_i$. For special dibits-$\Delta\theta_i$ assignment, such as the one for $\pi/4$–QPSK, as will be seen in the next section, the angle detector can be replaced by two threshold detectors (see Figure 4.31). The bit error probability of the suboptimum demodulator

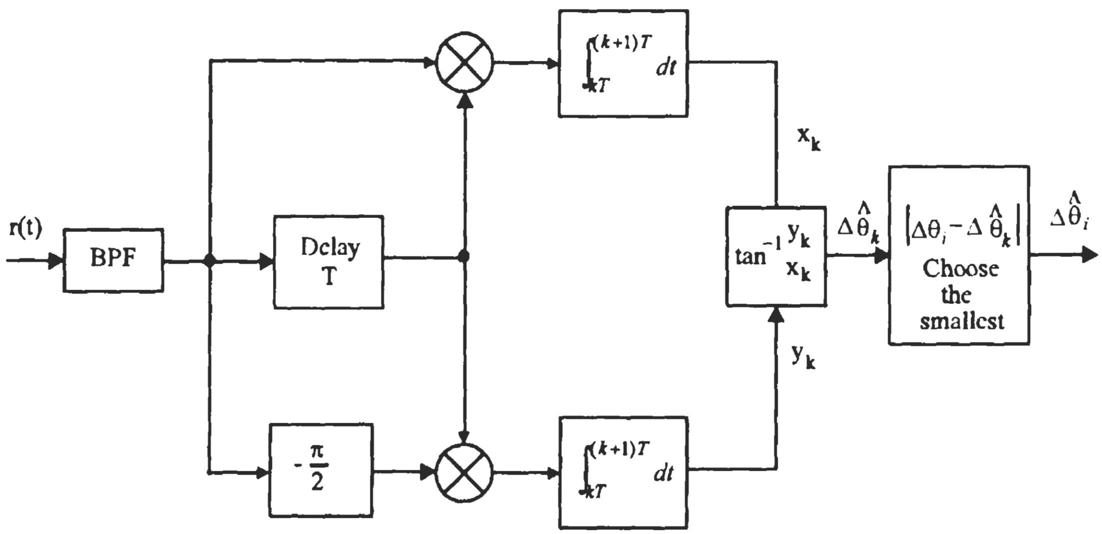


Figure 4.24 Suboptimum DQPSK demodulator (see Figure 4.16 for optimum demodulator).

in Figure 4.24 is given by [7, p.260]

$$P_b \approx e^{-(A^2/2\sigma^2)(1-1/\sqrt{2})} \tag{4.42}$$

where $A^2/2\sigma^2$ is the carrier-to-noise power ratio. In order to compare this to other error probabilities, we need to relate $A^2/2\sigma^2$ to E_b/N_o. In the derivation of this expression [7], the narrow-band noise has a variance of σ^2 for the inphase and quadrature component at the output of the front-end bandpass filter. The total noise variance is also equal to σ^2 [8, p.76]. The baseband signal is bandlimited to B. The bandwidth of the bandpass filter is just the same. Thus there is no intersymbol interference and the signal amplitude at sampling instances is A. So far in this chapter, the baseband pulse shape is always assumed as rectangular. For this pulse shape, the intersymbol interference free filter is the Nyquist filter which has a bandwidth of $B = 1/T$ at carrier frequency. Thus the noise power $\sigma^2 = N_o/T$. The signal symbol energy $E_s = \frac{1}{2}A^2T$. Thus

$$A^2/2\sigma^2 = \frac{2E_s/T}{2N_o/T} = \frac{E_s}{N_o} = \frac{2E_b}{N_o}$$

This is also stated in [2, p.444, eqn.(7.6)]. Thus (4.42) can be written as

$$P_b \approx e^{-\frac{2E_b}{N_o}(1-1/\sqrt{2})} = e^{-0.59\frac{E_b}{N_o}}, \text{(Suboptimum DQPSK)} \tag{4.43}$$

This is plotted in Figure 4.21. Seen from the figure, the degradation to the optimum

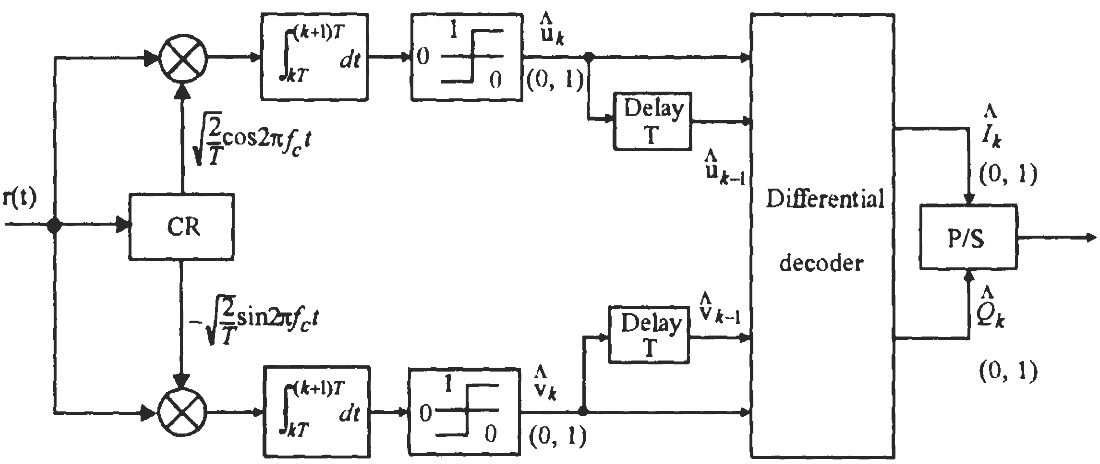


Figure 4.25 Coherent demodulation of DEQPSK.

DQPSK is less than 1 dB for high SNRs (>12 dB). But at the lower SNRs, the degradation is much larger. This is intuitively convincing since the reference signal is the previous signal which has noise. Compared to the suboptimum DBPSK, the degradation is about 1.75 dB for all SNRs.

For the purpose of phase ambiguity elimination, the DEQPSK signals are coherently demodulated. The demodulator is thus basically the same as QPSK demodulator except that a differential decoder must be inserted after demodulation. The demodulator can be in the form of Figure 4.17 where differential decoding is carried out on signal phases. However, since there are only two levels in the I- and Q-channels, the demodulator can be in a simpler form as shown in Figure 4.25, where the differential decoding is carried out on digital signal levels. The decoding rules are

$$\begin{aligned} \widehat{I}_k &= \overline{(\widehat{u}_k \oplus \widehat{v}_k)}(\widehat{u}_k \oplus \widehat{u}_{k-1}) + (\widehat{u}_k \oplus \widehat{v}_k)(\widehat{v}_k \oplus \widehat{v}_{k-1}) \\ \widehat{Q}_k &= \overline{(\widehat{u}_k \oplus \widehat{v}_k)}(\widehat{v}_k \oplus \widehat{v}_{k-1}) + (\widehat{u}_k \oplus \widehat{v}_k)(\widehat{u}_k \oplus \widehat{u}_{k-1}) \end{aligned} \tag{4.44}$$

Table 4.5 shows the decoding process assuming a phase ambiguity of $\pi/2$.

Note that the coherent demodulator for DEQPSK in Figure 4.25 is not suitable for $\pi/4-$QPSK since its dibits-$\Delta\theta_i$ assignment is different (see the $\pi/4-$QPSK section).

The symbol error probability has been given in (4.32). For Gray coded constel-

Transmitted absolute phases		$\frac{\pi}{4}$	$\frac{7\pi}{4}$	$\frac{\pi}{4}$	$\frac{7\pi}{4}$	$\frac{\pi}{4}$	$\frac{5\pi}{4}$	$\frac{3\pi}{4}$	$\frac{3\pi}{4}$	$\frac{7\pi}{4}$
Demodulation										
Estimated absolute phases		$\frac{3\pi}{4}$	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	$\frac{\pi}{4}$	$\frac{3\pi}{4}$	$\frac{7\pi}{4}$	$\frac{5\pi}{4}$	$\frac{5\pi}{4}$	$\frac{\pi}{4}$
Detected digits	$\widehat{u}_k$	0	1	0	1	0	1	0	0	1
	$\widehat{v}_k$	1	1	1	1	1	0	0	0	1
Detected information digits	$\widehat{I}_k$		1	0	1	0	1	1	0	1
	$\widehat{Q}_k$		0	1	0	1	1	0	0	1

Table 4.5 Differential decoding for DEQPSK.

lation and at high SNR, this translates to a bit error probability of

$$P_b \approx 2Q\left(\sqrt{\frac{2E_b}{N_o}}\right), \text{ (DEQPSK)} \tag{4.45}$$

which is plotted in Figure 4.21. It is seen from the figure that DQPSK is less than 0.5 dB inferior to coherent QPSK.

4.8 OFFSET QPSK

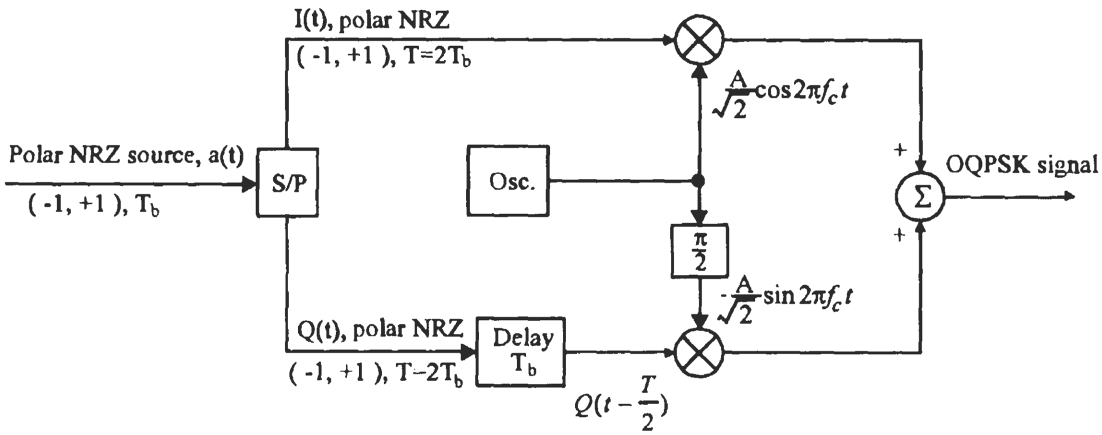
Offset QPSK is essentially the same as QPSK except that the I- and Q-channel pulse trains are staggered. The modulator and the demodulator of OQPSK are shown in Figure 4.26, which differs from the QPSK only by an extra delay of $T/2$ seconds in the Q-channel. Based on the modulator, the OQPSK signal can be written as

$$s(t) = \frac{A}{\sqrt{2}}I(t)\cos 2\pi f_c t - \frac{A}{\sqrt{2}}Q(t-\frac{T}{2})\sin 2\pi f_c t, \quad -\infty < t < \infty$$

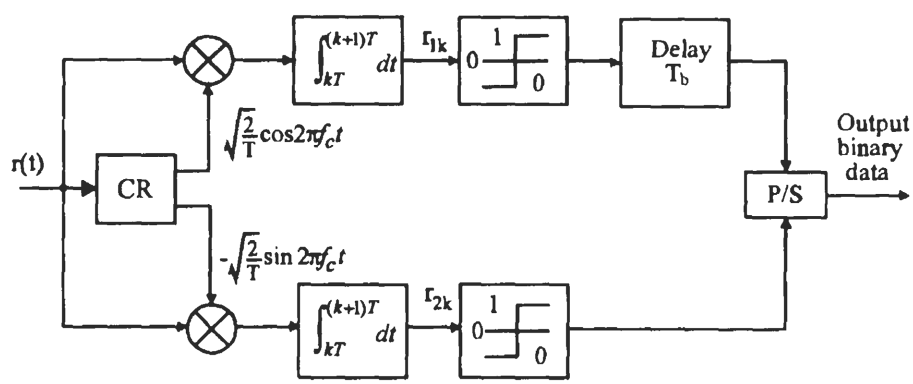
Since OQPSK differs from QPSK only by a delay in the Q-channel signal, its power spectral density is the same as that of QPSK, and its error performance is also the same as that of QPSK.

The OQPSK waveforms are shown in Figure 4.27. We observe that due to the staggering of I- and Q-channels, the OQPSK signal has a symbol period of $T/2$. At any symbol boundary, only one of the two bits in the pair (I_k, Q_k) can change sign. Thus the phase changes at symbol boundaries can only be $0°$ and $\pm 90°$. Whereas the QPSK signal has a symbol period of T, both two bits in the pair (I_k, Q_k) can change sign, and the phase changes at the symbol boundaries can be $180°$ in addition to $0°$and $\pm 90°$ (see Figure 4.19).

In comparison to QPSK, OQPSK signals are less susceptible to spectral side-lobe restoration in satellite transmitters. In satellite transmitters, modulated signals



(a)



(b)

Figure 4.26 OQPSK modulator and demodulator.

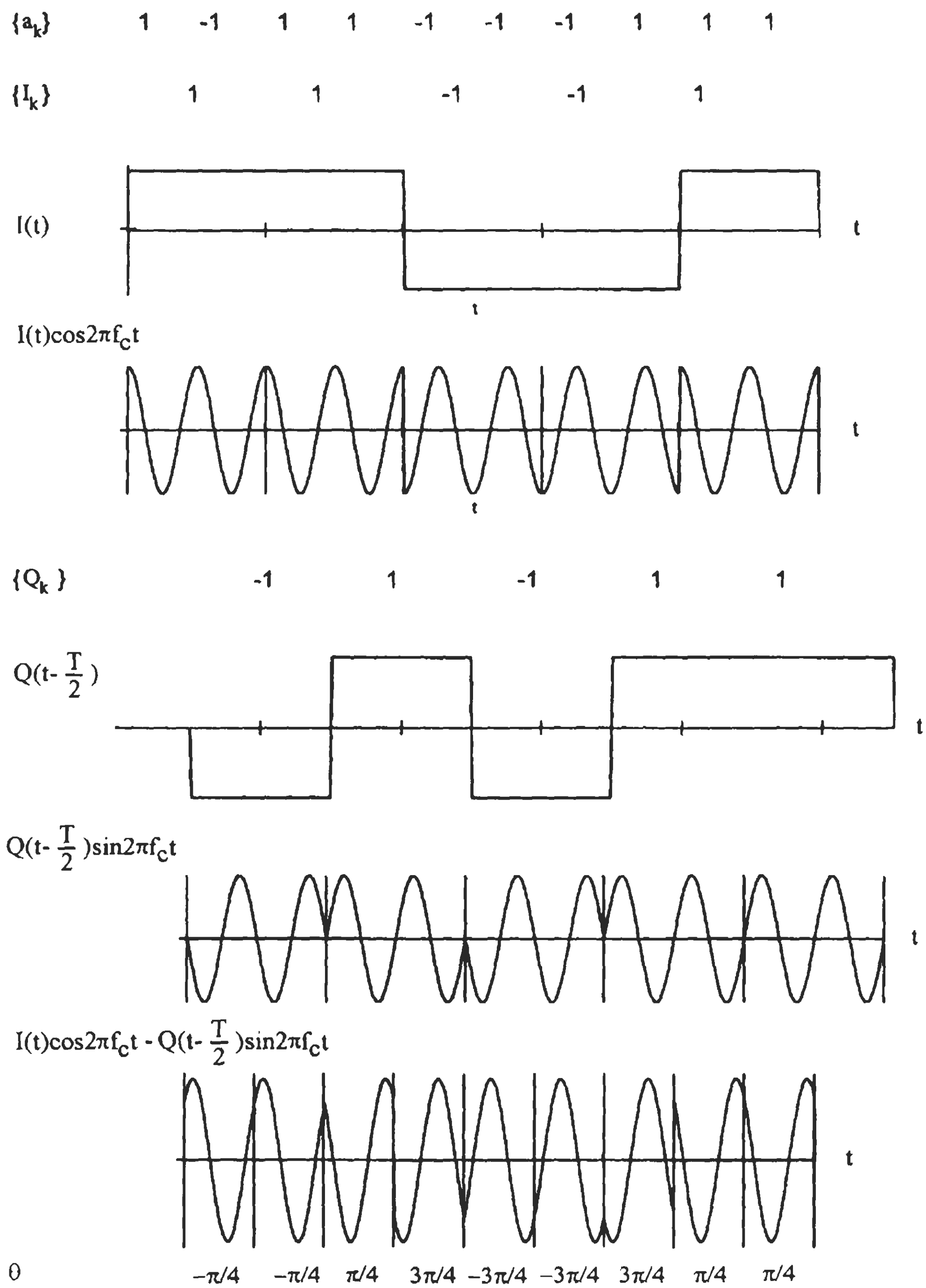


Figure 4.27 OQPSK waveforms.

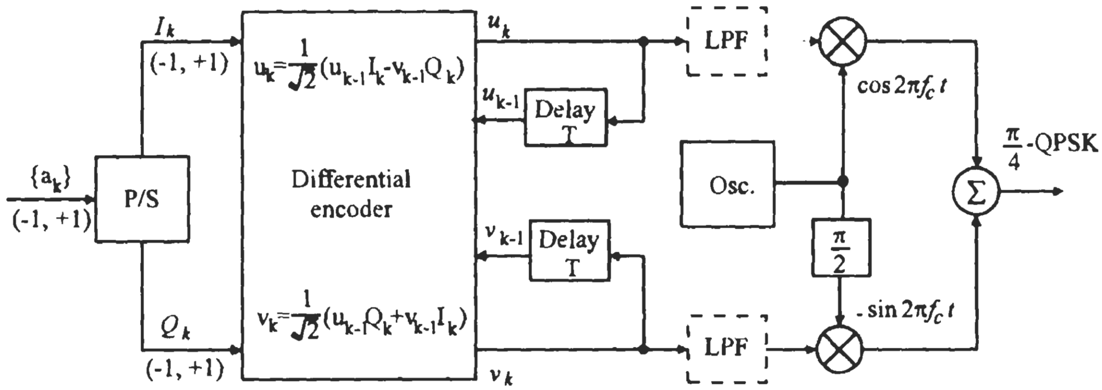


Figure 4.28 $\pi/4$-QPSK modulator.

must be bandlimited by a bandpass filter in order to conform to out-of-band emission standards. The filtering degrades the constant-envelope property of QPSK, and the 180° phase shifts will cause the envelope to go to zero momentarily. When this signal is amplified by the final stage, usually a highly nonlinear power amplifier, the constant envelope will be restored. But at the same time the sidelobes will be restored. Note that arranging the bandpass filter after the power amplifier is not feasible since the bandwidth is very narrow compared with the carrier frequency, the Q-value of the filter must be extremely high such that it cannot be implemented by the current technology. In OQPSK, since the 180° phase shifts no longer exist, the sidelobe restoration is less severe [9].

4.9 $\pi/4$-QPSK

Although OQPSK can reduce spectral restoration caused by nonlinearity in the power amplifier, it cannot be differentially encoded and decoded. $\pi/4$-QPSK is a scheme which not only has no 180° phase shifts like OQPSK, but also can be differentially demodulated. These properties make it particularly suitable to mobile communications where differential demodulation can reduce the adversary effect of the fading channel. $\pi/4-$ QPSK has been adopted as the standard for the digital cellular telephone system in the United States and Japan.

$\pi/4$-QPSK is first introduced by Baker in 1962 [10] and studied in [11, 12] and other articles.

The $\pi/4$-QPSK is a form of differentially encoded QPSK. But it differs from the DEQPSK described in the previous section by the differential coding rules. Figure 4.28 is the $\pi/4$-QPSK modulator. $(I(t), Q(t))$ and $(u(t), v(t))$ are the uncoded and coded I-channel and Q-channel bits. The differential encoder of $\pi/4$-QPSK modulator encodes $I(t)$ and $Q(t)$ into signals $u(t)$ and $v(t)$ according to the following rules

$$\left.\begin{array}{l} u_k = \frac{1}{\sqrt{2}}(u_{k-1}I_k - v_{k-1}Q_k) \\ v_k = \frac{1}{\sqrt{2}}(u_{k-1}Q_k + v_{k-1}I_k) \end{array}\right\} \tag{4.46}$$

where u_k is the amplitude of $u(t)$ in the kth symbol duration and so on. We assume that I_k, Q_k takes values of $(-1, 1)$. If we initially specify that $u_0 = 1$ and $v_0 = 0$, then u_k and v_k can take the amplitudes of $\pm 1, 0$, and $\pm 1/\sqrt{2}$. The output signal of the modulator is

$$\begin{aligned} s(t) &= u_k \cos 2\pi f_c t + v_k \sin 2\pi f_c t \\ &= A\cos(2\pi f_c t + \Phi_k), \quad kT \le t \le (k+1)T \end{aligned}$$

where

$$\Phi_k = \tan^{-1}\frac{v_k}{u_k}$$

which depends on the encoded data, and

$$A = \sqrt{u_k^2 + v_k^2}$$

is independent of time index k, that is, the signal has a constant envelope. This can be easily verified by substituting (4.46) into the expression of A, it turns out $A_k = A_{k-1}$. In fact $A = 1$ for initial values $u_0 = 1$ and $v_0 = 0$. It can be proved that the phase relationship between two consecutive symbols is

$$\left.\begin{array}{l} \Phi_k = \Phi_{k-1} + \Delta\theta_k \\ \Delta\theta_k = \tan^{-1}\frac{Q_k}{I_k} \end{array}\right\} \tag{4.47}$$

where $\Delta\theta_k$ is the phase difference determined by input data.

Proof: By definition

$$\begin{aligned} \tan\Phi_k &= \frac{v_k}{u_k} \\ &= \frac{u_{k-1}Q_k + v_{k-1}I_k}{u_{k-1}I_k - v_{k-1}Q_k} = \frac{Q_k + \frac{v_{k-1}}{u_{k-1}}I_k}{I_k - \frac{v_{k-1}}{u_{k-1}}Q_k} \\ &= \frac{Q_k + \tan\Phi_{k-1}I_k}{I_k - \tan\Phi_{k-1}Q_k} = \frac{Q_k\cos\Phi_{k-1} + I_k\sin\Phi_{k-1}}{I_k\cos\Phi_{k-1} - Q_k\sin\Phi_{k-1}} \end{aligned}$$

$I_k Q_k$	$\Delta\theta_k$	$\cos\Delta\theta_k$	$\sin\Delta\theta_k$
1 1	$\pi/4$	$1/\sqrt{2}$	$1/\sqrt{2}$
-1 1	$3\pi/4$	$-1/\sqrt{2}$	$1/\sqrt{2}$
-1 -1	$-3\pi/4$	$-1/\sqrt{2}$	$-1/\sqrt{2}$
1 -1	$-\pi/4$	$1/\sqrt{2}$	$-1/\sqrt{2}$

Table 4.6 $\pi/4$-QPSK signal phase assignment.

Now let

$$\begin{aligned} I_k &= \sqrt{2}\cos\Delta\theta_k \\ Q_k &= \sqrt{2}\sin\Delta\theta_k \end{aligned} \tag{4.48}$$

then

$$\Delta\theta_k = \tan^{-1}\frac{Q_k}{I_k}$$

and we have

$$\begin{aligned} \tan\Phi_k &= \frac{\sin\Delta\theta_k\cos\Phi_{k-1} + \cos\Delta\theta_k\sin\Phi_{k-1}}{\cos\Delta\theta_k\cos\Phi_{k-1} - \sin\Delta\theta_k\sin\Phi_{k-1}} \\ &= \frac{\sin(\Phi_{k-1}+\Delta\theta_k)}{\cos(\Phi_{k-1}+\Delta\theta_k)} = \tan(\Phi_{k-1}+\Delta\theta_k) \end{aligned}$$

thus we have proved (4.47). Using (4.48) we can write (4.46) as

$$\begin{aligned} u_k &= u_{k-1}\cos\Delta\theta_k - v_{k-1}\sin\Delta\theta_k \\ v_k &= u_{k-1}\sin\Delta\theta_k + v_{k-1}\cos\Delta\theta_k \end{aligned} \tag{4.49}$$

Table 4.6 shows how $\Delta\theta_k$ is determined by the input data.

Referring to the values of $\Delta\theta_k$ in Table 4.6, we can see clearly from (4.47) that the phase changes are confined to odd-number multiples of $\pi/4$ (45°). There are no phase changes of 90° or 180°. In addition, information is carried by the phase changes $\Delta\theta_k$, not the absolute phase Φ_k. The signal constellation is shown in Figure 4.29. The angle of a vector (or symbol) with respect to the positive direction of axis u is the symbol phase Φ_k. The symbols represented by • can only become symbols represented by $\times$, and vice versa. Transitions among themselves are not possible. The phase change from one symbol to the other is $\Delta\theta_k$.

Since information is carried by the phase changes $\Delta\theta_k$, differentially coherent demodulation can be used. However, coherent demodulation is desirable when higher power efficiency is required. There are four ways to demodulate a $\pi/4$–QPSK

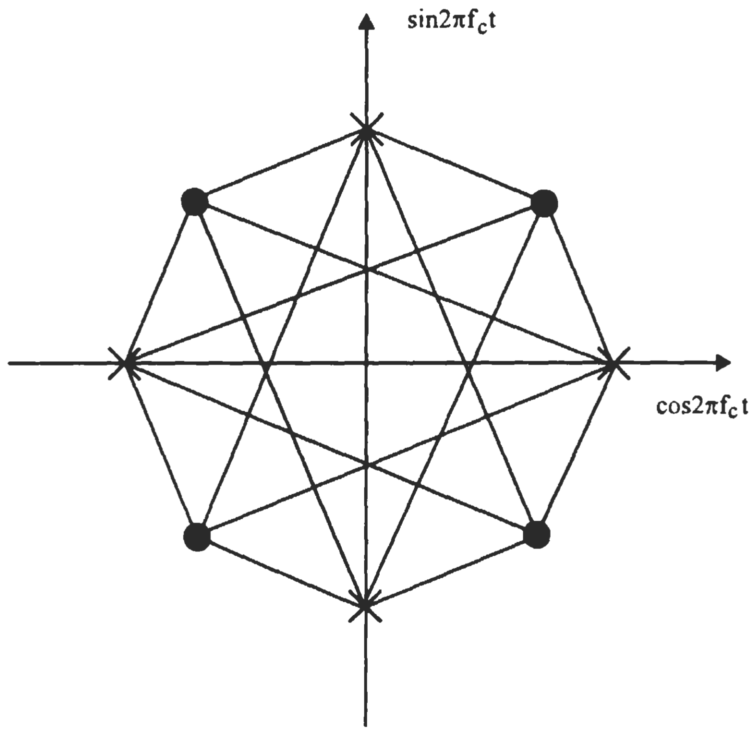


Figure 4.29 $\pi/4$-QPSK signal constellation.

signal:[6]

1. Baseband differential detection,
2. IF band differential detection,
3. FM-discriminator detection,
4. Coherent detection.

The first three demodulators are reported to be equivalent in error performance [11]. The coherent demodulator is 2 to 3 dB better.

Figure 4.30 is the baseband differential demodulator which is just a special case of the DMPSK demodulator in Figure 4.16. The LPF in Figure 4.30 is equivalent to the integrator in Figure 4.16. The angle calculation and comparison stages in Figure 4.16 are equivalently replaced by two threshold detectors. The bandpass filter (BPF) at the front end is used to minimize the noise power. However, carrier phase must be preserved for the proper differential detection. A square-root raised-cosine roll-off BPF can achieve this goal [12]. The local oscillation has the same frequency as the

6 As a matter of fact, these methods are also applicable to other differential MPSK schemes.

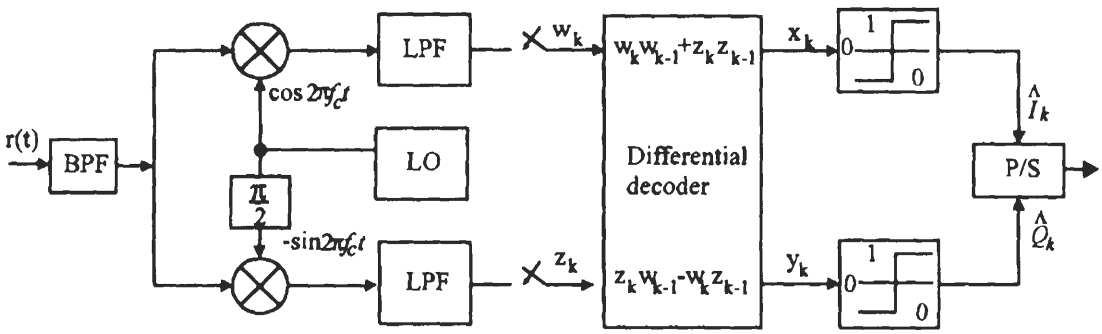


Figure 4.30 Baseband differential demodulator for $\pi/4$-QPSK.

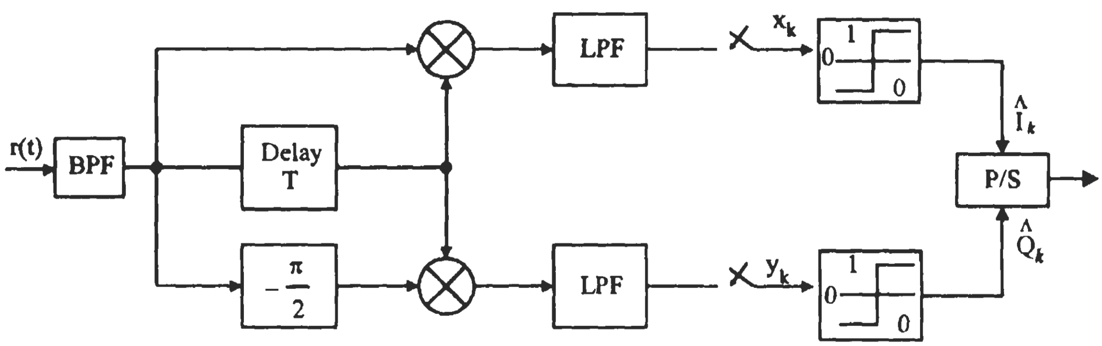


Figure 4.31 IF band differential demodulator for $\pi/4$-QPSK.

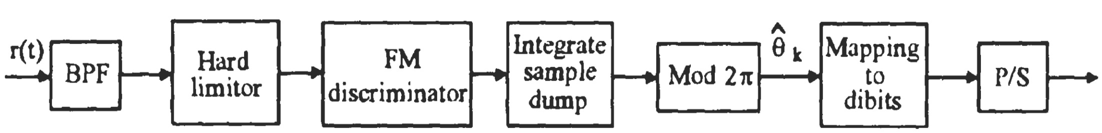


Figure 4.32 FM-discriminator demodulator for $\pi/4$-QPSK.

unmodulated carrier but its phase is not synchronous with the received signal. It is assumed that the difference phase θ in the received signal remains essentially the same from the previous symbol duration to the current symbol duration. This phase difference will be cancelled in the baseband differential decoder.

In the absence of noise, the output of the BPF in the kth symbol duration is

$$r(t) = A_k \cos(2\pi f_c t + \Phi_k + \theta), \qquad kT \le t \le (k+1)T \tag{4.50}$$

where θ is the random phase introduced by the channel. We assume that θ changes very slowly in comparison to the symbol rate so that it is considered constant in two consecutive symbols. The time-varying amplitude A_k has replaced the constant amplitude in the transmitted signal. The variation in amplitude might be due to channel fading or interference. In the kth symbol duration the I-channel multiplier output is

$$A_k \cos(2\pi f_c t)\cos(2\pi f_c t + \Phi_k + \theta) = \frac{1}{2}A_k\left[\cos(4\pi f_c t + \Phi_k + \theta) + \cos(\Phi_k + \theta)\right]$$

The low-pass filter (LPF) output for the I-channel is therefore (ignoring the factor $1/2$ and the LPF loss)

$$w_k = A_k \cos(\Phi_k + \theta)$$

Similarly the Q-channel LPF output is

$$z_k = A_k \sin(\Phi_k + \theta)$$

Since θ has not been changed from the previous symbol duration, then

$$w_{k-1} = A_{k-1}\cos(\Phi_{k-1} + \theta)$$

$$z_{k-1} = A_{k-1}\sin(\Phi_{k-1} + \theta)$$

The decoding rule is

$$\begin{aligned} x_k &= w_k w_{k-1} + z_k z_{k-1} \\ y_k &= z_k w_{k-1} - w_k z_{k-1} \end{aligned}$$

which is

$$\begin{aligned} x_k &= A_k A_{k-1}\left[\cos(\Phi_k + \theta)\cos(\Phi_{k-1} + \theta) + \sin(\Phi_k + \theta)\sin(\Phi_{k-1} + \theta)\right] \\ &= A_k A_{k-1}\cos(\Phi_k - \Phi_{k-1}) = A_k A_{k-1}\cos\Delta\theta_k \\ y_k &= A_k A_{k-1}\left[\sin(\Phi_k + \theta)\cos(\Phi_{k-1} + \theta) - \cos(\Phi_k + \theta)\sin(\Phi_{k-1} + \theta)\right] \\ &= A_k A_{k-1}\sin(\Phi_k - \Phi_{k-1}) = A_k A_{k-1}\sin\Delta\theta_k \end{aligned}$$

From Table 4.6, the decision devices decide[7]

$$\begin{aligned}\widehat{I}_k &= 1, \text{if } x_k > 0 \text{ or } \widehat{I}_k = -1, \text{if } x_k < 0 \\ \widehat{Q}_k &= 1, \text{if } y_k > 0 \text{ or } \widehat{Q}_k = -1, \text{if } y_k < 0\end{aligned}$$

The symbol error probability is given in (4.29) or (4.30), and P_b is given in (4.40).

The IF band differential demodulator (Figure 4.31) cancels the phase difference θ in the IF band. The I-channel multiplier output is

$$\begin{aligned}& A^2 \cos(2\pi f_c t + \Phi_k + \theta)\cos(2\pi f_c t + \Phi_{k-1} + \theta) \\ = \; & \frac{1}{2}A^2 \left[(\cos(4\pi f_c t + \Phi_k + \Phi_{k-1} + 2\theta) + \cos(\Phi_k - \Phi_{k-1})\right]\end{aligned}$$

Again ignoring the factor $A^2/2$ and the LPF loss, LPF output is $\cos(\Phi_k - \Phi_{k-1}) = x_k$. Similarly the Q-channel LPF output is found to be $\sin(\Phi_k - \Phi_{k-1}) = y_k$. The rest is the same as the baseband differential detection. The advantage of this demodulator is that no local oscillator is needed.

The discriminator demodulator is shown in Figure 4.32. The ideal bandpass hard limiter keeps the envelope of the received signal constant without changing its phase. The ideal frequency discriminator output is proportional to the instantaneous frequency deviation of the input signal. That is

$$v(t) = \frac{d}{dt}\left(\Phi(t) + \theta\right)$$

The integrate-sample-dump (ISD) circuit output is

$$\begin{aligned}\int_{(k-1)T}^{kT} v(t)dt &= \Phi_k - \Phi_{k-1} \\ &= \Delta\theta_k + 2n\pi \\ &= \Delta\theta_k, (\text{mod } 2\pi)\end{aligned}$$

where $2n\pi$ is caused by click noise [13]. The modulo-2π operation removes the $2n\pi$ term and the output is $\Delta\theta_k$ which in turn will be mapped to a corresponding dibit.

The error probabilities of the above three demodulators are reported to be equal [11].[8] Like the DQPSK in the last section, the symbol error probability for the above three equivalent demodulators is given by (4.29) or (4.30). The bit error probability

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[7] In Figure 4.16, the decision rule is to choose the smallest $|\Delta\theta_i - \Delta\widehat{\theta}_k|$. This rule is simplified for $\pi/4$-QPSK as described here.

[8] But it is not clear why the IF band differential demodulator could be equivalent to the baseband differential demodulator, because the latter is optimum and the former is not, according to our discussion in the section discussing DQPSK. The same doubt should arise regarding the discriminator demodulator.

can be approximately calculated using (4.25). Alternately, bit error probability can be evaluated using (4.40).

The coherent $\pi/4$-QPSK demodulator can be in the form of Figure 4.17 where differential decoding is performed on the signal phases. The coherent demodulator of DEQPSK in Figure 4.25 is not suitable for $\pi/4$-QPSK since its dibits-$\Delta\theta_i$ assignment is different. A novel coherent $\pi/4$-QPSK demodulator (Figure 4.33) has been proposed in [11] where differential decoding is performed on the baseband signal levels. In $\pi/4$-QPSK, assuming $A_k = 1$ in (4.50), the demodulated signals are two-level $(\pm 1/\sqrt{2})$ at every other sampling instant. In between, the signals are three-level $(0, \pm 1)$. This can be seen from the signal constellation (Figure 4.29), where $\bullet$ signals are two-level and $\times$ signals are three-level, and a $\bullet$ signal must be followed by a $\times$ signal or vice versa. If three-level detection is employed, the performance degrades compared with two-level detection. The three-level signals are converted to two-level signals in Figure 4.33. The converted two-level signals are detected by a two-level threshold detector. When the signals are two-level, the switches are in position A, the detection is the same as in QPSK. When the signals are three-level, the switches are in position B, the signals are converted to two-level by the following simple operations

$$\begin{aligned} x_{2k} &= x_{1k} - y_{1k} \\ y_{2k} &= x_{1k} + y_{1k} \end{aligned} \tag{4.51}$$

It is easy to verify that the conversions are (ignoring noise)

$$\begin{array}{ccc} (x_{1k}, y_{1k}) & \longrightarrow & (x_{2k}, y_{2k}) \\ (+1, 0) & \longrightarrow & (+1, +1) \\ (-1, 0) & \longrightarrow & (-1, -1) \\ (0, +1) & \longrightarrow & (-1, +1) \\ (0, -1) & \longrightarrow & (+1, -1) \end{array}$$

This is equivalent to rotate vector (x_{1k}, y_{1k}) by $+\pi/4$ and amplify its amplitude by $\sqrt{2}$. In other words, it is to rotate a $\times$ vector to the next $\bullet$ vector position with an amplitude gain of $\sqrt{2}$. This makes signal power doubled. However, the noise power is also doubled since the in-phase and quadrature channel noise are uncorrelated (see (4.51)). Thus the BER performance of the coherent $\pi/4$-QPSK is the same as that of the coherent QPSK.

The detected signals $\widehat{u}_k$ and $\widehat{v}_k$ are decoded by a DEQPSK differential decoder as in (4.44). Then the signals must be passed through a special P/S converter. The circuit is shown in Figure 4.34. The clock is derived from the symbol clock by dividing the frequency by two. The phase of this clock is synchronous to the switch

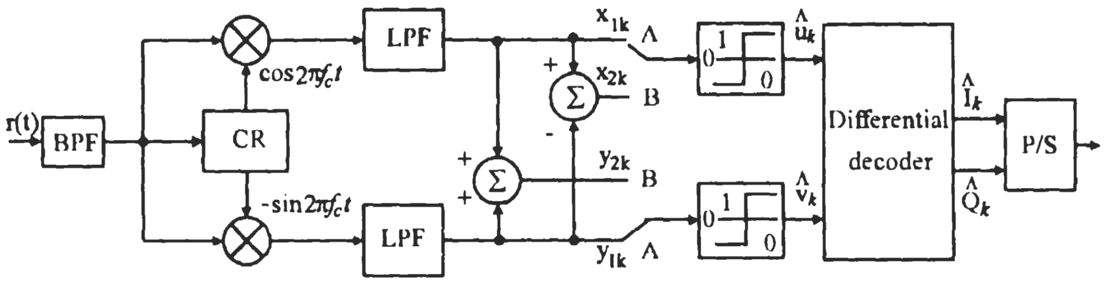

Figure 4.33 $\pi/4$-QPSK coherent demodulator.

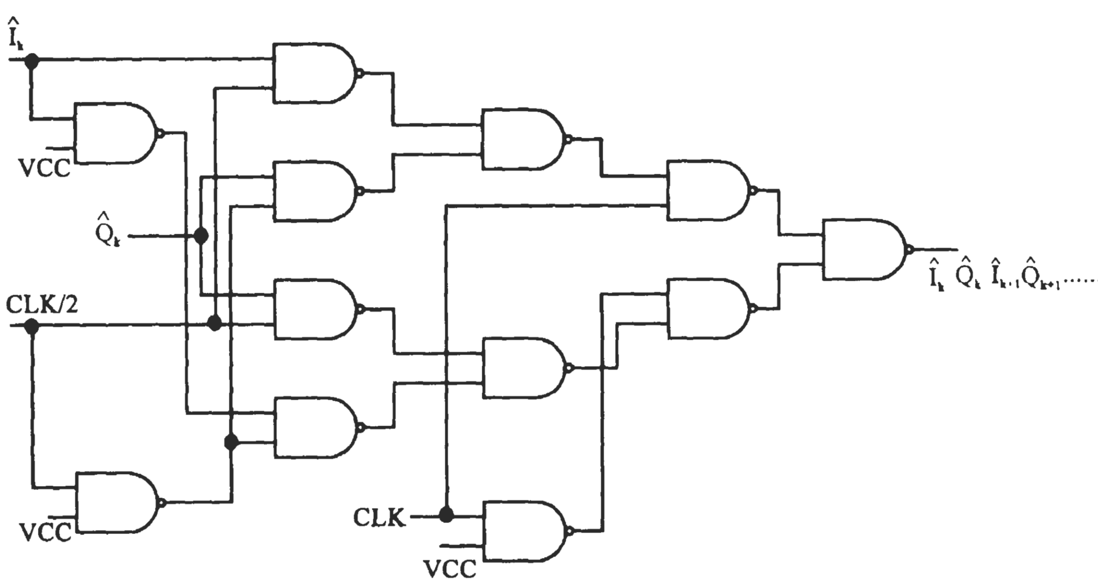

Figure 4.34 The parallel-to-serial converter of the $\pi/4$-QPSK coherent demodulator.

for selecting sampled signals. The S/P converter delivers inphase and quadrature data alternately to the output. Refer to [11] for details.

As mentioned above in terms of error probability in the AWGN channel, the $\pi/4$-QPSK is the same as ordinary DEQPSK or DQPSK. That is, the coherently demodulated $\pi/4$-QPSK has the same BER as that of DEQPSK and the noncoherently demodulated $\pi/4$-QPSK has the same BER as that of DQPSK (see Figure 4.21).

4.10 SYNCHRONIZATION

Coherent demodulation requires that the reference signal at the receiver be synchronized in phase and frequency with the received signal. Both coherent and noncoherent demodulations require symbol timing at the receiver to be synchronized in phase and frequency with the received signal.

Carrier synchronization can be achieved by sending a pilot tone before message signals. Because the pilot tone has a strong spectral line at the carrier frequency, the receiver can easily lock on it and generates a local coherent carrier. However, this requires extra transmission bandwidth.

Carrier synchronization also can be achieved with a carrier recovery circuit which extracts the phase and frequency information from the noisy received signal and use it to generate a clean sinusoidal reference signal.

Symbol synchronization usually is achieved by a clock (symbol timing) recovery circuit which uses the received signal to control the local oscillator.

4.10.1 Carrier Recovery

The PSK signals have no spectral line at carrier frequency. Therefore a nonlinear device is needed in the carrier recovery circuit to generate such a line spectrum. There are two main types of carrier synchronizers, the Mth power loop, and the Costas loop.

Figure 4.35 is the Mth power loop for carrier recovery for M-ary PSK. For BPSK (or DEBPSK), $M = 2$, thus it is a squaring loop. For QPSK (or OQPSK, DEQPSK), $M = 4$, it is a quadrupling loop, and so on. It is the Mth power device that produces the spectral line at Mf_c. The phase lock loop consisting of the phase detector, the LPF, and the VCO, tracks and locks onto the frequency and phase of the Mf_c component. The divide-by-M device divides the frequency of this component to produce the desired carrier at frequency f_c and with almost the same phase of the received signal. Before locking, there is a phase difference in the received signal relative to the VCO output signal. We denote the phase of the received signal as θ and the phase of the VCO output as $M\widehat{\theta}$.

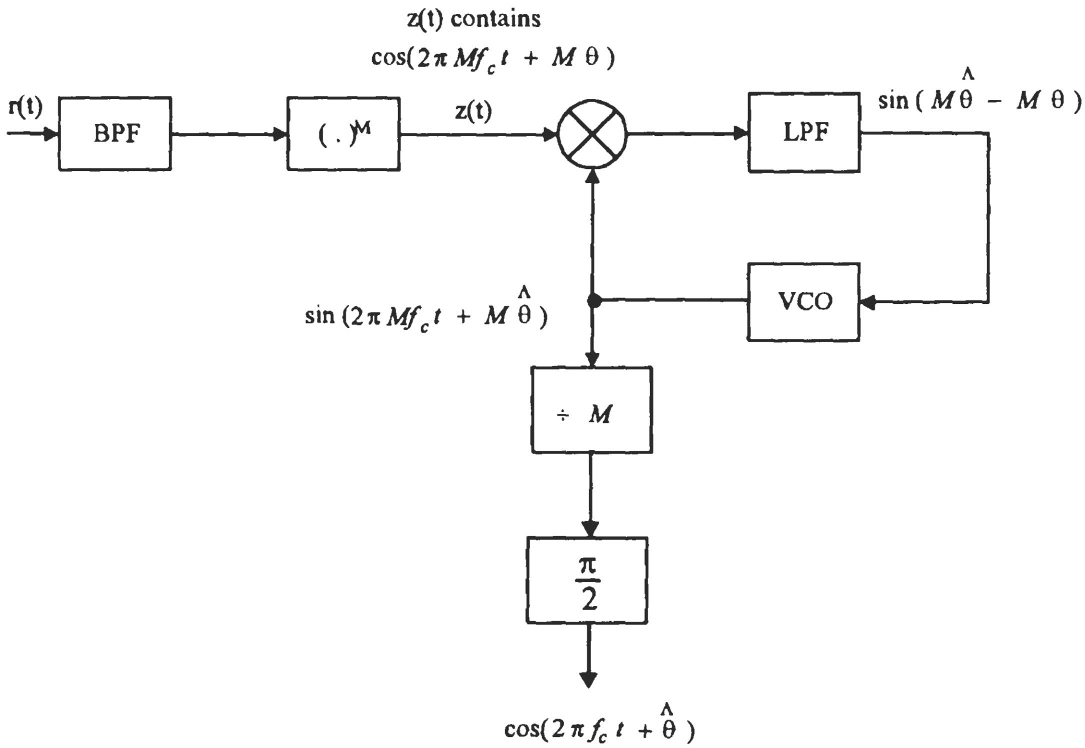

Figure 4.35 Mth power synchronizer for carrier recovery.

For BPSK, using (4.5), setting $A = 1$, and noticing that $a^2(t) = 1$, we have

$$\begin{aligned} z(t) &= [s(t) + n(t)]^2 = a^2(t)\cos^2(2\pi f_c t + \theta) + \text{ noise terms} \\ &= \frac{1}{2}[1 + \cos(4\pi f_c t + 2\theta)] + \text{noise terms} \end{aligned}$$

which contains a spectral line at $2f_c$ that can be tracked by the phase lock loop (PLL). The VCO output is divided by two in frequency to provide the desired carrier. It is obvious that the loop will produce a carrier with the same phase when the phase θ is either 0 or π. Then the demodulator output could be $+a(t)$ or $-a(t)$. We say that the loop has a phase ambiguity of π. Differential coding can eliminate phase ambiguity, as we described in previous sections in this chapter.

For QPSK, using (4.35), setting $A = \sqrt{2}$, and noticing that $I^2(t) = Q^2(t) = 1$, we have

$$\begin{aligned} z(t) &= [s(t) + n(t)]^4 \\ &= \{[I(t)\cos(2\pi f_c t + \theta) - Q(t)\sin(2\pi f_c t + \theta) + n(t)]^2\}^2 \end{aligned}$$

$$= \quad [1 - I(t)Q(t)\sin(4\pi f_c t + 2\theta)]^2 + \text{noise terms}$$
$$= \quad 1 - 2I(t)Q(t)\sin(4\pi f_c t + 2\theta) + \frac{1}{2} - \frac{1}{2}\cos(8\pi f_c t + 4\theta) + \text{noise terms}$$

The last signal term contains a spectral line at $4f_c$ which is locked onto by the PLL. A divide-by-four device is used to derive the carrier frequency. Note that the $I(t)Q(t)\sin(4\pi f_c t + 2\theta)$ term resulted from squaring operation cannot produce a line spectrum since $I(t)Q(t)$ has a zero mean value. Therefore fourth power operation is needed for QPSK (and OQPSK). The last term will have a 0 initial phase for $\theta = 0$, or $\pm\pi/2$. The demodulator output could be $\pm I(t)$ or $\pm Q(t)$. This is to say that there is a $\pi/2$ phase ambiguity in the carrier recovery. It can be eliminated by differential coding as we discussed before.

For general MPSK, where $M = 2^n$, the Mth power operation will produce a spectral line at Mf_c and the phase ambiguity is $2\pi/M$.

The performance of the Mth power loop is generally measured by the phase tracking error. Under the usual small angle approximation (i.e., the phase error is small so that $M\widehat{\theta} - M\theta \approx \sin(M\widehat{\theta} - M\theta)$), the variance of such error is given by [14]

$$\sigma_\theta^2 = 2N_o B_L S_L \tag{4.52}$$

where B_L is the loop bandwidth defined in terms of the loop transfer function $H(f)$ as follows

$$B_L = \int_0^\infty |H(f)|^2 df, \quad \text{(Hz)}$$

The parameter S_L is the upper bound of squaring loss in the BPSK case and quadrupling loss in the QPSK case. S_L is a number without unit that reflects the increase in the variance of phase error due to squaring or quadrupling operation in the phase tracking loop. They are given by [14]

$$S_L = 1 + \frac{1}{2\rho_i}, \text{ for } M = 2$$

and

$$S_L = 1 + \frac{9}{\rho_i} + \frac{6}{\rho_i^2} + \frac{3}{2\rho_i^3}, \text{ for } M = 4$$

where ρ_i is the input signal-to-noise ratio of the carrier recovery circuit. Observing (4.52), the unit of σ_θ^2 is seemingly watt instead of radian^2. This is due to the small-angle approximation in deriving the phase error variance where volt is replaced by radian. Therefore the unit of σ_θ^2 should be radian^2.

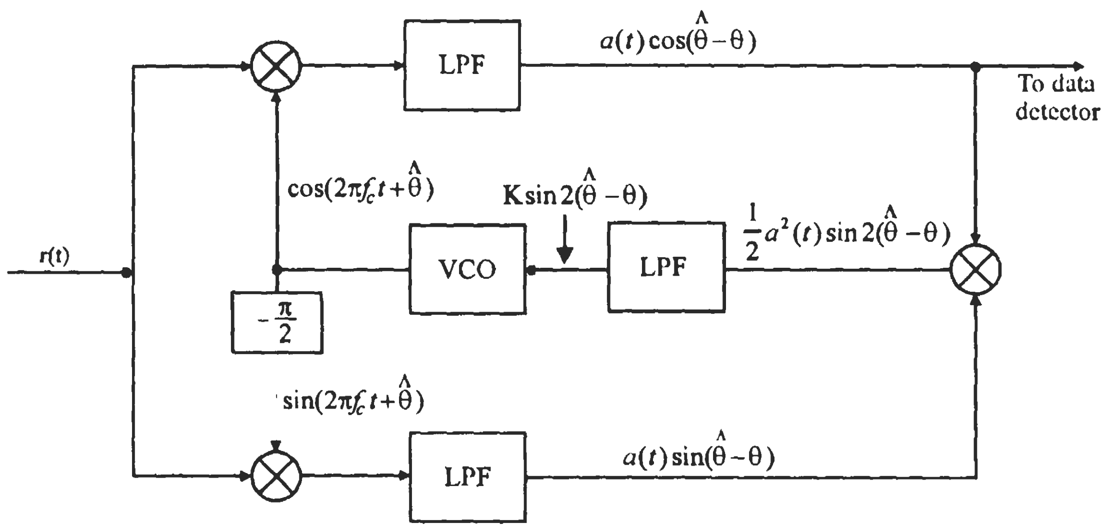

Figure 4.36 Costas loop for carrier recover for BPSK.

A difficulty in circuit implementation of the Mth power loop is the Mth power device, especially at high frequencies. Costas loop design avoids this device.

Figure 4.36 is the Costas loop for carrier recovery for BPSK. Initially the VCO generates a sinusoid with a frequency close to the carrier frequency f_c and some initial phase. The frequency difference and the initial phase are accounted for by the phase $\widehat{\theta}$. The multipliers in the I- and Q-channels produce $2f_c$ terms and zero frequency terms. The LPFs attenuate the $2f_c$ terms and their outputs are proportional to $a(t)\cos(\widehat{\theta}-\theta)$ or $a(t)\sin(\widehat{\theta}-\theta)$. Then these two terms multiply again to give the term $\frac{1}{2}a^2(t)\sin 2(\widehat{\theta}-\theta)$ which is low-pass filtered one more time to get rid of any amplitude fluctuation in $a^2(t)$, thus the control signal to the VCO is proportional to $\sin 2(\widehat{\theta}-\theta)$, which drives the VCO such that the difference $\widehat{\theta}-\theta$ becomes smaller and smaller. For sufficiently small $\widehat{\theta}-\theta$, the I-channel output is the demodulated signal.

The Costas loop for QPSK is shown in Figure 4.37. The figure is self-explanatory and its working principle is similar to that of BPSK. The limiters are bipolar, which are used to control the amplitude of the two channels' signal to maintain balance. When the phase difference $\phi = \widehat{\theta}-\theta$ is sufficiently small, the I- and Q-channel outputs are the demodulated signals.

A difficulty in Costas loop implementation is to maintain the balance between the I- and Q-channel. The two multipliers and low-pass filters in these two channels must be perfectly matched in order to achieve the theoretical performance.

Although the appearance of the Mth power loop and the Costas loop are quite

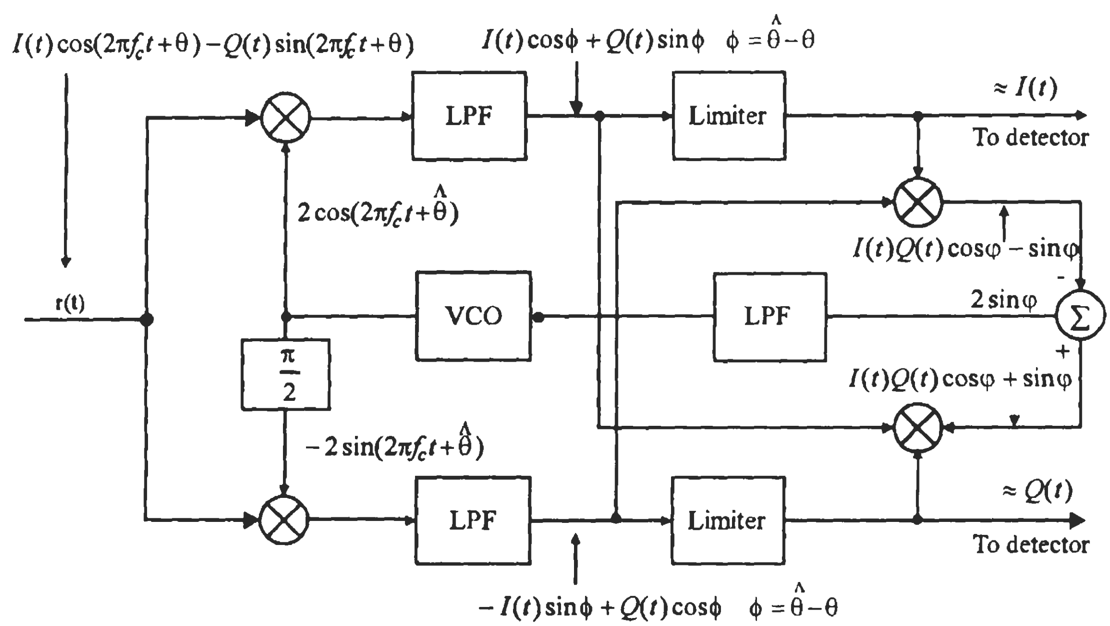

Figure 4.37 Costas loop for carrier recovery for QPSK.

different, their performance can be shown to be the same [14]. Therefore (4.52) is also applicable to Costas loops.

4.10.2 Clock Recovery

The clock or symbol timing recovery can be classified into two basic groups. One group is the open loop synchronizer which uses nonlinear devices. These circuits recover the clock signal directly from the data stream by nonlinear operations on the received data stream. Another group is the closed-loop synchronizers which attempt to lock a local clock signal onto the received data stream by use of comparative measurements on the local and received signals.

Two examples of the open-loop synchronizer are shown in Figure 4.38. The data stream that we use in the phase shift keying modulation is NRZ waveform. Recall in Chapter 2 we have shown that this waveform has no spectral energy at the clock frequency (see Figure 2.3(a)). Thus in the open-loop synchronizers in Figure 4.38, the first thing that one needs to do is to create spectral energy at the clock frequency. In the first example, a Fourier component at the data clock frequency is generated by the delay-and-multiply operation on the demodulated signal $m(t)$. This frequency component is then extracted by the BPF that follows and shaped into square wave by the final stage. The second example generates the clock frequency

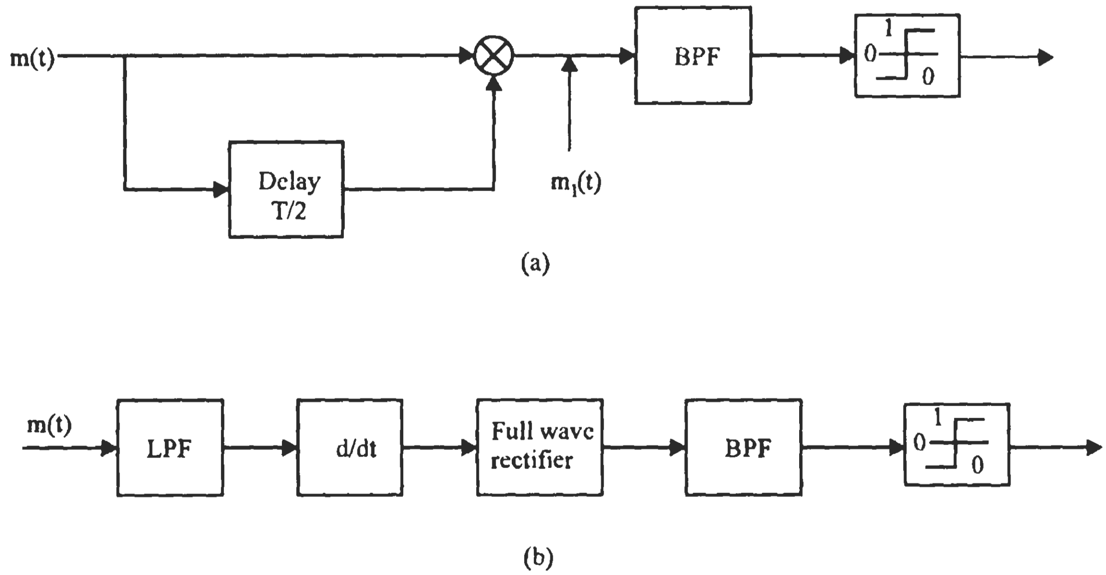

Figure 4.38 Two types of open-loop symbol synchronizers.

component by using the differentiator-rectifier combination. The differentiator is very sensitive to wideband noise, therefore a low-pass filter is placed in the front end of the synchronizer.

An early/late-gate circuit shown in Figure 4.39 is an example of the class of closed-loop synchronizers. The working principle is easily understood by referencing Figure 4.40. The time zero point is set by the square wave clock locally generated by the VCO. If the VCO square wave clock is in perfect synchronism with the demodulated signal $m(t)$, the early-gate integrator and the late-gate integrator will accumulate the same amount of signal energy so that the error signal $e = 0$. If the VCO frequency is higher than that of $m(t)$, then $m(t)$ is late by $\Delta < d$, relative to the VCO clock. Thus the integration time in the early-gate integrator will be $T - d - \Delta$, while the integration time in the late-gate integrator is still the entire $T - d$. The error signal will be proportional to $-\Delta$. This error signal will reduce the VCO frequency and retard the VCO timing to bring it back toward the timing of $m(t)$. If the VCO frequency had been lower and the timing had been late, the error signal would be proportional to $+\Delta$, and the reverse process would happen, that is, the VCO frequency would be increased and its timing would be advanced toward that of the incoming signal.

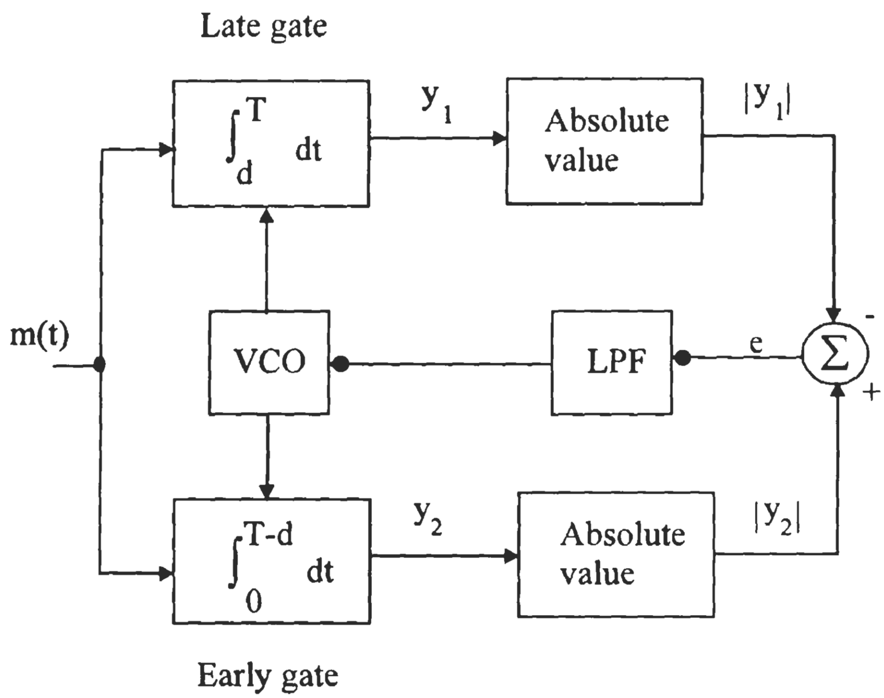

Figure 4.39 Early/late-gate clock synchronizer.

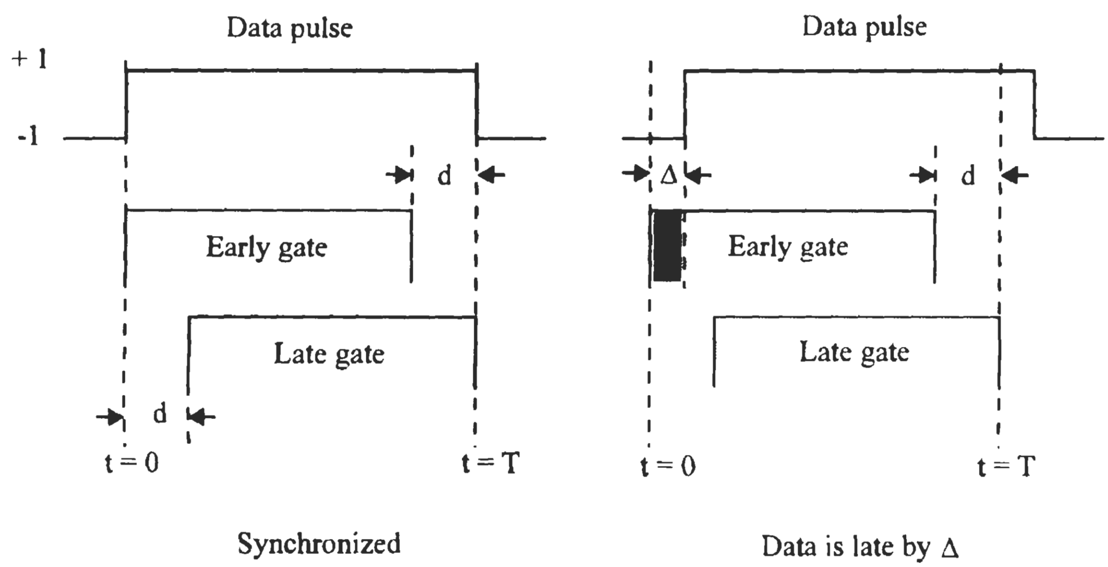

Figure 4.40 Early/late-gate timing illustration.

4.10.3 Effects of Phase and Timing Error

As an example, we want to check the effects of phase and timing error on the bit error probability of coherent BPSK in the AWGN channel.

If the local carrier is in error by ϕ radians, the correlator output amplitude will be reduced by a factor $\cos\phi$. Thus the conditional bit error probability of BPSK will be

$$P_b(\phi) = Q\left(\sqrt{\frac{2E_b}{N_o}}\cos\phi\right) \tag{4.53}$$

If ϕ is Gaussianly distributed with variance σ_ϕ^2, the bit error probability is [15, p.270-271]

$$\begin{aligned} P_b &= \int_{-\infty}^{\infty} p(\phi)P_b(\phi)d\phi \\ &= \frac{2}{\sqrt{2\pi}\sigma_\phi}\int_0^{\infty}\exp\left(-\frac{\phi}{2\sigma_\phi^2}\right)Q\left(\sqrt{\frac{2E_b}{N_o}}\cos\phi\right)d\phi \end{aligned} \tag{4.54}$$

This is plotted in Figure 4.41 where different curves are labeled according to values of the standard deviation of the phase error σ_ϕ in radians. It is seen that when $\sigma_\phi < 0.2$, the degradation is not significant.

The effect of a symbol synchronization error on the bit error probability of BPSK depends on the presence or absence of a symbol transition. If two successive symbols are identical, an incorrect symbol reference will have no effect on the error probability. If two successive symbols differ, the magnitude of the correlator output is reduced by a factor of $1-(2|\Delta|/T)$ where Δ is the timing error. Thus given a timing error, the conditional P_b is

$$\begin{aligned} P_b(\Delta) &= \Pr(\text{error} \mid \text{transition}, \Delta)\Pr(\text{transition} \mid \Delta) \\ &\quad + \Pr(\text{error} \mid \text{no transition}, \Delta)\Pr(\text{no transition} \mid \Delta) \end{aligned}$$

If the successive symbols are independent and equally likely to be either of the two binary symbols, the probability of transition is one-half, and, if the normalized timing error $\tau = \Delta/T$ is Gaussianly distributed, then

$$\begin{aligned} P_b &= \frac{1}{2\sqrt{2\pi}\sigma_\tau}\int_{-0.5}^{0.5}\exp\left(-\frac{\tau}{2\sigma_\tau^2}\right)Q\left(\sqrt{\frac{2E_b}{N_o}}(1-2|\tau|)\right)d\tau \\ &\quad +\frac{1}{2}Q\left(\sqrt{\frac{2E_b}{N_o}}\right) \end{aligned} \tag{4.55}$$

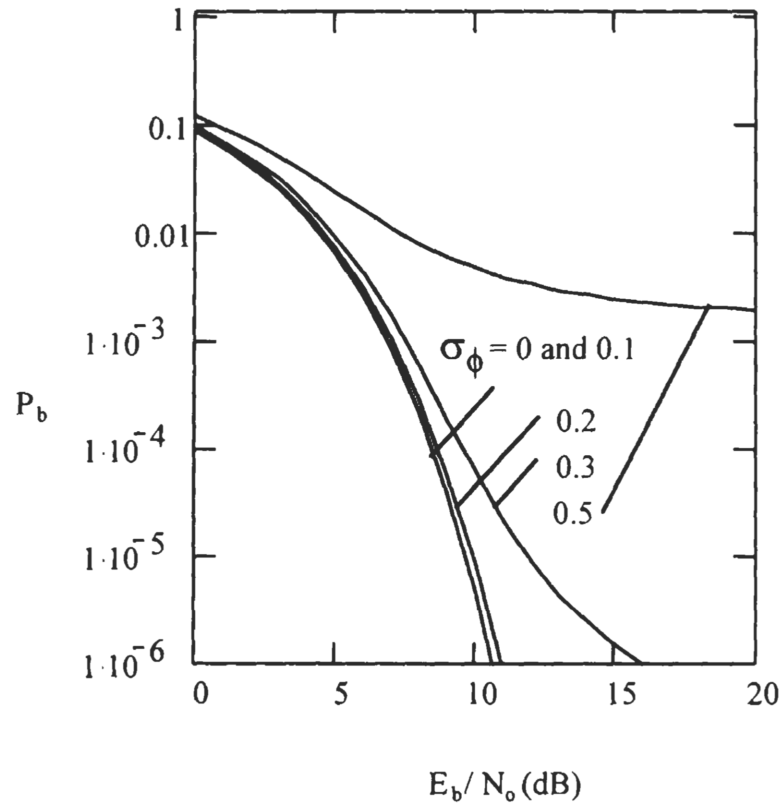

Figure 4.41 Effect of imperfect carrier synchronization on P_b of BPSK.

where the integration limits are 0.5 to –0.5 which is the range of the timing errors [16. section 6-4 and 9-4]. This is plotted in Figure 4.42, where different curves are labeled according to a value of the $\pi\sigma_\tau$ where σ_τ is the standard deviation of the timing error τ. When $\pi\sigma_\tau$ is less than 0.2, the degradation is not significant.

It should be pointed out that (4.54) and (4.55), consequently Figure 4.41 and 4.42, are based on the assumption that the errors are Gaussianly distributed [15. p.270-271], which may not be accurate, depending on the carrier and clock recovery systems. More accurate, but also more complicated, error distribution models and error probability results are given in [16, section 6-4 and 9-4].

4.11 SUMMARY

In this chapter we have covered all important PSK modulation schemes. We de-